

**SOCIAL MEDIA AND VACCINE HESITANCY IN THE CONTEXT OF COVID-19 IN
ZIMBABWE**

by

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ABSTRACT

The study analyses the ways in which social media (WhatsApp, Twitter, Facebook and YouTube) information contributes to Covid-19 vaccine hesitancy in Zimbabwe in order to establish how digital media is be used as a propaganda campaign against formulated ways to fight against the ravaging and lethal coronavirus. The study was premised on the fact that social media has become so influential in the fast dissemination of information, whether true or false, and thus shapes public opinion within a short space of time. Therefore, the study specifically examines the nature of the messages circulated on social media platforms concerning vaccination analysing the possible impact it had on vaccine uptake. The research advanced the contention that the impact of the messages posted with regard to vaccination should be understood from the way(s) it was framed by the digital media. Its communication through the media affected its meaning as it was mediated by media spectacle, fake news, myths regarding the vaccine, and the implications posed by sending messages in a comical or satirical way. The study also established the ways in which digital communication in the form of social media platforms mentioned above can be used for wrong reasons in spite of its benefits to society and also advocates for the use of social media responsibly in ways that uphold the respect human rights on both online and offline. It was argued that social media in Zimbabwe played a crucial role in shaping people's perceptions towards vaccination against Covid-19 since anti-vaccine messages were implicated in fake news, conspiracy theories, misinformation and disinformation.

DEDICATION

This research is dedicated:

to my mother, Esther Muwandi.

to my wife, Privilege Chikono.

to my late father, Heater Zvidzai.

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I would like to acknowledge the contributions of the people who assisted me in a variety of ways as I was doing this research. Firstly, I want to thank my supervisor Professor M. Musiyiwa who tirelessly supported me from commencement to until the end. I owe him an intellectual debt for his constant guidance all over the project.

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I want to thank the Almighty God, for his word in Psalms 33:9 was fulfilled. It is also important to highlight the contribution made by my lecturer at the University of Zimbabwe, Dr R. Marongedze among others.

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CHAPTER ONE

INTRODUCTION

1.1 Area of Investigation

This study investigates how social media contributes to the problem of Covid-19 vaccine hesitancy in Zimbabwe in order to establish how digital media is being used as a propaganda campaign against strategies to combat the highly lethal coronavirus. Whereas scientists repeatedly advise that inoculation is the most effective preventive measure against hospitalization and/or death due to Covid-19 infection, in Zimbabwe and the rest of the world, large populations have shunned vaccination. This can be seen from the statistics governments provide of citizens who have received at least one jab of a vaccine. Besides, while governments and health concerned non-governmental organisations (NGOs) are relentlessly campaigning spreading information to help individuals and communities combat the novel coronavirus disease, there are also Covid-19 denialists and skeptics spreading misinformation and disinformation against the current WHO guidelines and protocols, including vaccination, to fight the pandemic (Musiyiwa, 2022). When the pandemic started and vaccines later introduced, social media platforms in Zimbabwe and across the globe became awash with anti-Covid-19 and counter-inoculation information. There is little doubt that such information contributed to vaccine hesitancy defined by MacDonald et al (2015) as “the deferral in the acceptance and at the same time rejection of vaccines despite the availability of provided vaccination services.” The latest statistics regarding Covid-19 vaccination in Zimbabwe state that 6, 327, 092 people have received the first dose, 4, 677, 607 the second dose, 903, 634 the booster shot. The total of fully vaccinated population so far is therefore 4, 621, 086 people. According the 2002 Census prepared by Central Statistical Office (CSO), Zimbabwe has a population of 11, 631, 657 people. According to current projections, Zimbabwe is estimated to have 15, 298, 328 people. From February 2022 the government rolled out another vaccination campaign that included children from 12 years and above after the Paediatric Association of Zimbabwe, which brings together specialists in child health, gave the greenlight. Whereas the target population has been expanded and 31,1 percent of the target population vaccinated, a large number of people have shunned

Covid-19 vaccination. It is estimated that 70.3 percent of the population is still unvaccinated. The question to be asked is: Why are they still unvaccinated in spite of clear evidence many people dying of the pandemic? There should be compelling reasons, though medically ill-informed, why such people refuse inoculation. A cursory look at social media platforms such as WhatsApp, YouTube, Twitter and Facebook shows large amounts of anti-vaccination information, textual, visual or both are being circulated. This study thus intends to access that information and analyse it in order to establish why it is so influential in persuading many people against vaccination. Such an establishment will help to understand how social media can be used responsibly in contexts that present communities with serious challenges and the need for collective action to confront such challenges.

1.2 Statement of Problem

Digital technology has been very instrumental in the local and global efforts to combat the Covid-19 pandemic since its inception in China in November 2019. A pandemic is an epidemic that has spread over several countries or continents, impacting many people. Since 2020 with the first coronavirus vaccine, Sputnik V, being scientifically proven as effective in fighting the virus through preventing severe illness, hospitalisation or death, vaccination was embraced globally by governments and private health players to fight the pandemic. The world's most information dissemination tool, digital media, became an avenue to propagate the importance and benefits of vaccination in the wake of exponential infections and deaths from the virus. At the same time it became a platform used by anti-vaccination and corona virus denialists and skeptics to spread inoculation misinformation and disinformation bordering on conspiracy theories. With mass demonstrations against vaccination, lockdowns, Covid-19 passes, isolation and other WHO approved measures to fight Covid-19, it shows that vaccine hesitancy is very high locally and globally. This study intends to analyse how digital platforms, WhatsApp, Facebook and YouTube were used to propagate information promoting vaccine hesitancy in Zimbabwe (and globally) with the sole purpose of establishing how digital technology can be used for the wrong reasons - disrupting strategies currently in use to fight the pandemic thus prolonging any chances to eradicate the virus and prevent illness and death.

1.3 Aims of the study

The study aims to establish the ways in which digital communication in the form of social media can be used for wrong reasons in spite of its benefits to society. It also aims to advocate for the use of social media responsibly in ways that uphold the respect of others' (human) rights on both online and offline.

1.4 Research Objectives

- (i) To identify anti-Covid-19 vaccination information posted on the selected social media platforms.
- (ii) To explain the ways in which such misinformation promotes vaccine hesitancy in Zimbabwe.
- (iii) To establish how digital spaces can be used for wrongs which complicate instead of contributing to addressing societal challenges.

1.5 Research Questions

- (i) What is vaccine hesitancy?
- (ii) How has social media been used to disseminate anti-inoculation messages?
- (iii) What kind of anti-vaccination information has been posted on social media platforms selected for study?
- (iv) How influential has been this information in promoting supporting conspiracy theories and precipitating vaccine hesitancy?
- (v) How can social media be used responsibly for the development of society?

1.6 Research Justification

The significance of this study is derived from the fact that it adds to a growing corpus of literature that focus on COVID-19 vaccine hesitancy. The uniqueness of the current stems from its pursuit to proffer a digital perspective through the use of digital tools. To be more frankly, it

provides a cyber-geospatial perspective, which plays a crucial role in understanding how the pandemic has unfolded over space and time on cyber space specifically on social media platforms. A nommoic approach is also going to be utilised in analysing the effect of the messages send on different social media platforms. Furthermore, the study is worth pursuing because researchers so far have mainly studied the COVID-19 vaccine hesitancy in Zimbabwe outside the purview of the utilisation of digital tools for analysis.

The choice of focusing on factors behind coronavirus vaccine hesitancy is not from the impression that the researcher is the only one who wrote about vaccine hesitancy in Zimbabwe, but the contemporary study is aware of others such as Chiyaka, Mukora-Mutseyekwa and Makurumidze among others. What makes this research distinct from these is that this research adopts both of traditional and digital equipment for data analysis.

This research also marked as a great contribution to the country's educational sector because it is in line with Education 5.0 (a policy introduced in Zimbabwe that includes practical based learning). As such, this study becomes a potentially reliable imaginative resource for explicating the researcher's motivation behind social media and COVID-19 vaccination.

1.7 Methodology

This study is a mixed or quali-quantitative research that relies on qualitative and quantitative methods of gathering, presenting, sampling and analysing data. Some mixed methods researchers consider this form of research a methodology and focus on the philosophical assumptions. The approach is necessitated by the fact that Digital Humanities is a multidisciplinary discipline. The mixed methods research approach underscores the employment of several. It focuses on mixing both of quantitative and qualitative information in a particular research. Mixed method is a kind of research with theoretical assumptions along with quantitative and qualitative methods. According to (Brannen1992:7), 'Here the researcher's choice of methods is said to be chiefly driven by the philosophical assumptions that is epistemological and ontological. Mixed method encompasses philosophical expectations that directs the collection and examination of data. "...mixed methods research is an opportunity that deflects attention away from theoretical work

that is often specific to particular disciplines. Thus it may encourage thinking ‘outside the box’, a practice to be welcomed.” (Brannen, 1992)

The central principle in mixed methods research is the utilisation of the two approaches in a combination because it provides a better understanding of the given research problems as compared to a singular approach. The quantitative research includes information that is limited (close-ended) such as that found on performance instruments. Astalin (2013: 1) describes a qualitative research as ‘a systematic scientific inquiry which seeks to build a holistic, largely narrative, description to inform the researcher’s understanding of a social or cultural “phenomenon”.’ This reveals that qualitative research pivots on narrative and descriptive analysis of data. The study purposively selects WhatsApp, Twitter, Facebook and YouTube as digital platforms for gathering covid-19 vaccination information.

In presenting data, the study uses the summation method in which textual scope of messages are summarised and punctuated with illustrative excerpts extrapolated from the social media information. The summation method is explained by dictionary.com as a method for associating a sum with a divergent series. In analysing primary data, the research employs nommoic creativity as a method of analysing short texts. Nommoic creativity is a rhetorical discourse approach that focuses on the generative and creative power of the word (Asante, 1998 and Ani, 1980). The generative and creative power of these words in syntactic choreographed patterns becomes the entry point for this study’s analytical engagement. The frames in this case, text messages under analysis influence campaign effectiveness in promoting coronavirus disease 2019 (COVID-19) vaccines. It is important to understand how social media as an information source influences vaccination campaign effectiveness as regards the attempt to fight the Covid-19 disease.

This research is also going to make use a descriptive cross-sectional survey using a self-administered interview among adults. To substantiate the claims drawn from the primary sources, the study carried out semi-structured interviews with different Zimbabwean citizens. The interviews will assess those people who were willing to get vaccinated, distinctive attitudes and perceptions among individuals regarding the effectiveness and efficacy of the vaccine. It is

also going to make use of secondary sources that analyse vaccine hesitancy in general and coronavirus vaccine hesitancy in particular. Some of these sources include hard and soft copy books, and the conventional media (newspapers), popular songs and videos.

Under quantitative research, the researcher is going to use Mozdeh Software for gathering social media (Facebook, Twitter; and YouTube) information regarding vaccine hesitancy specifically focusing on the causes of vaccine hesitancy, and people's reaction to hesitancy. The researcher is also going to adopt Social Media Indicator analysis (SMI). This (SMI) is a useful tool in indicating the extent of engagement of those people who were against the vaccine on social media platforms and the impact of their messages. SMI is explained by Liberto (2021) as a social sentiment that analyses aggregated social media data to help organisations understand how they are performing in the eyes of consumers.

1.8 Theoretical Framework

This study adopts a duplex approach blending Media spectacle and Framing theories by Guy Debord and Erving Goffman respectively. The 'framing' concept was firstly posited by Gregory Bateson (1972). Relying on psychological frames, he defined them as a "spatial and temporary set of cooperative messages" (Bateson, 1972, p. 197) that function as a mode of metacommunication (Hallahan, 2008). Entman (1993:51) refers to 'framing' as "a dispersed conceptualisation"; with whose background studies lack strong intangible definitions and relying on context-specific, instead of generally appropriate operationalisations. Erving Goffman in 1974 recovered the concept of 'framing' and he further perpetuated it for the field of sociology, and it was this renewed sociological perspective was applied in communication studies. According to him, a frame is a social framework and mental schema that lets users to organise experiences. The framing theory conceptualizes information being given to the audience as influential to the choices they make and how they react to that information. According to Olasunkanmi (2017:1), "Frames conveyed by the news or media place is the most common use of frames. With regard to social media, the framing theory suggests that social media generate this kind of frame by presenting news objects using pre-defined contextualisation.

Communication scholars normally use the term “frame” in two ways (Druckman, 2001; Scheufele, 1999). Firstly, a media frame refers to the used words, images, phrases, and presentation styles by a speaker when presenting information about an argument to an audience. Secondly, a personal frame refers to an individual’s intellectual understanding of a given situation (Goffman, 1974). When some of social media users engage with the issue of vaccines they deal with contradictory type of frames that are anticipated by the leaders of the opinion to influence communal general preferences. While on the one hand there are frames used by pro-vaccination campaigners, there are also influential opposing frames used by anti-vaccination campaigners thereby causing vaccine hesitancy.

Framing in this case focuses on how cyberspace attracts the society’s eye to major trending issues and then create a frame, which the spectators will assimilate such information. The Mass Communication Theory of 2017 assumes that, the foundation of frames is that media focuses consideration on common occasions and then places them inside an arena of meaning. Public attitude oftenly depends on how the society chooses to frame basic issues in a society. For instance, citizens’ ideas regarding the Covid-19 vaccination program depend on whether people frame the vaccine as an adequate anti-viral medicine or a toxin that will worsen the situation. Explaining Framing concept, Goffman (1974), puts argues that, ‘...publics interpret what is going on in their society through their own framework’. In the case of coronavirus, the messages about the advancement of any vaccine was conveyed using different media “frames” that were applied in any message.

There are various framing methods of vaccines, ‘positive’ and ‘negative’ are the two main ones. In these two framing could also include religious, political, scientific/medical, economic, human rights and other factors. The image below is a relevant example from a Twitter handle posted by a participant with the username Dr. Rae. The message is revealing concerning how Covid-19 vaccination was being framed as a useless antidote. In this case, the coronavirus vaccine is being framed negatively.



From the afore-displayed image, COVID-19 anti-vaccine messages are moderately prevalent in Zimbabwe, and social media in this case, Twitter, Facebook, WhatsApp and YouTube appears to be intensifying them. There is a high level of support against the herd immunity, this is revealed by the number of the likes (38) as compared to the dislikes. Immunity is the refusal by an organism a particular infection by the action of specific leucocytes (white blood cells). Mundagowa, Tozivepi, Chiyaka, Mukora-Mutseyekwa & Makurumidze (2021: 3) note that “A lot of people lacked trust in the Zimbabwean government and they began to interrogate vaccine effectiveness and efficacy.’ Several people believe conspiracy theories regarding Covid-19 vaccines but a remarkable smaller group do not, with belief especially high among younger people on social media (Duffy et al, 2020:8)

The Media Spectacle theory explains the notion of "society of the spectacle" coined by Guy Debord a French theorist,. The theory had a great impact on a variety of modern theories of society and culture. It refers to mediated events on digital platforms in which media forms like social media (Facebook, Twitter, WhatsApp, and YouTube, among others) process certain events in a spectacular form. Using the term media spectacle, Guy Debord largely focused on a range of systems of technically fabricated media productions that are created and circulated through mass media, ranging from traditional media to the Internet and the newest wireless gadgets.

Kellner (1995) explains further the concept of spectacle in relation to the media by stating that it means, ‘...media paradigms that are extraterrestrial and customary day-to-day routine which will become special media “spectacles”. Kellner builds on Debord’s conception of the society of spectacle but there is a tiny different. According to Debord (1967: 10), spectacle “unifies and explains a great diversity of apparent phenomena”. Examples of spectacularised events in

Zimbabwe include the vaccine media spectacle, Tinashe Mugabe DNA show, Vice President of Zimbabwe Kembo Mohadi's alleged unfaithful love scandals, Zimbabwe's controversial presidential elections of 2018, the formation of the central opposition party by Nelson Chamisa named Citizens Coalition for Change (CCC) in 2022, the ritual murder of Tawanda Makore of Murehwa, to mention but a few. The formation of CCC became a debatable topic on social media. There were those people that argue that CCC is a new political party and others arguing that it is still the Movement of Democratic Change - Alliance (MDC-A) that has merely altered its name. Due to this kind of debate, social media became a war fighting domain between different rivalries regarding this issue, hence it became a media spectacle. The same applies to those arguing for the existence of Covid-19 and the need to vaccinate the population against those denying the existence of the disease and the efficacy of vaccines. Due to digital media ordinary people are given opportunity to efficiently engage with information and discussions about Covid-19 vaccination. Media Spectacle can also sell a lie and news without facts what is typically called "fake news". The spread of fake news through various social media platforms has been increasing on a tremendous speed. Khan, Jain, Chouhan & Sheikh (2020: 365) postulate that;

...social media also allows the extreme production of "fake news". Fake news on social media have negative impacts that they pose in the society, thereby attracting great attention.

On social media, news on any topical issue such as the Covid-19 pandemic spread without any verification thereby disturbing the trust of the public as it tend to have faith in everything sent to them. Social media platforms has become popular for information consumption due to its low charges thus becoming the best media type for the prevalent of false information. The media spectacle theory will help establishing authenticity and falsehood in social media information about the Covid-19 vaccines in particular and the pandemic in general.

The mainstream shared media today in Zimbabwe processes news in the form of media spectacle. In an arena of massive race with daily television networks, radios, Internet sites and common new media which emerged due to "Web 2.0" like Facebook, WhatsApp, YouTube; and

Twitter consist of sensationalistic and tabloidised stories which they create in the forms of media spectacle that attempt to attract mega-audience until the next spectacle emerges. Web 2.0 is also called the people-centric Web or read and write Web. Murugesan (2007:34) states that;

Web 2.0 is the second phase in the internet's development and it is attracting the attention of Information Technology professionals, different businesses, and Web users... The Web 2.0 connects the Web in a more cooperative and collaborative manner and also presents new opportunities for leveraging the Web and allowing the users to engage more effectively.

Web 2.0 is an umbrella term encompassing several new social media networks that are essentially scrutinized by the so-called media spectacle. Media Spectacle is usually characterised by propaganda and alternative interpretation. Social media plays an important role in people's life with increasing media and cyber-culture making novel sites like Facebook, Twitter, WhatsApp; and YouTube. Zimbabwean people's social life is increasingly intermediated by media spectacle. Political, and socio-economic life is also shaped more by media spectacle. Social and political conflicts are increasingly shown on the displays of media culture, which exhibits spectacles like sex scandals by celebrities and politicians, and the volatile violence of day-to-day life.

Zimbabwe's approach to combating the coronavirus by adopting the vaccines made people's reactions on social media look absurd. The manifestations of forms of the hesitancy of Covid-19 vaccines were presented by a variety of critical works not only on vaccine hesitancy but also on acceptance of vaccine uptake and the effect of social media information. This section reviews critical works on its various perspectives as shown in diverse countries of the world regarding the issue of Covid-19 vaccine hesitancy. It also focuses on various critical works across Zimbabwe and Africa in general which have an impact on the audience and Zimbabwean community. Thus this study attends to various sources related to Covid-19 vaccination and it acts as a fundamental tool that is useful in the Zimbabwean community to acquire herd immunity for Covid-19.

1.9 Literature Review

Apuke and Omar (2021) demonstrated issues that caused fake news distribution throughout the Covid-19 health crisis using the affordance and cognitive load theories. Theoretically, their study develops the current study by concentrating on the the abundance of online information in forecasting fake news sharing behaviour among social media users, specifically in Nigeria. Evidence has exposed that the advent of COVID-19 as a global epidemic has brought with it an unprecedented amount of fake news which threatens global well-being. Epidemic refers to a sudden intensification in the number of cases of a certain disease, more than what is normally expected in a specific region. COVID-19 has reached epidemic proportions in China in mid-January 2020.

Mundagowa, Tozivepi, Chiyaka, Mukora-Mutseyekwa and Makurumidze (2021) assess the Covid-19 vaccine hesitancy among Zimbabweans. According to them, the vaccine uptake is hindered by the pervasive social media misrepresentation. People lacked trust in the government's ability to guarantee the availability of an effective vaccine. Zimbabwe targeted 60-72% of the total population.

Akther and Nur (2022) investigate the major factors influencing the acceptance of Covid-19 vaccines and they further build up a model grounded on the theory of rational action, belief in conspiracy theory, awareness, and perceived ease of use.

Bode (2015) reveal role of social media in disseminating political information. He specialises on political information occurs via Facebook and Twitter. He further suggests that social media plays a substantial role in the wisdom of political information within the modern media atmosphere.

Cook and Lewandowsky (2020) divulge the role of social media in amplifying the conspiracy theories. Social media has formed an arena in which any individual can reach as many people as other media. The lack of conventional justifiers is one reason why misinformation spreads faster online than true information, oftenly propelled by ghost accounts. Conspiracy theorists are more

disposed to like and share conspiracist posts on Facebook. Wood in Cook and Lewandowsky (2020) did an analysis of tweets about the Zika virus and there is a high rate of the propagators of conspiracy theories.

Chong and Druckman (2007) review the meaning of framing theory and the effects of frames on public opinion. After defining framing and framing effects, they articulate methods for identifying frames and the extent they affect the public. Chong and Druckman's study is quite theoretically useful to the current study which also examines how language is framed in the social media and the effect of that to communicating vaccine hesitancy.

Kellner (2009) exposes how certain events became media spectacles in the United States. He argues that, the typical media today in the United States process information in the form of media spectacle. In an arena of extreme competition with daily cable television networks, radio, Internet sites and ever proliferating social media sites like Facebook, MySpace, YouTube, and Twitter, competition for attention is more extreme because it is where a lot of information is spectacularised.

CHAPTER TWO

BACKGROUND TO THE RISE AND SPREAD OF COVID-19 AND THE NEED FOR VACCINATION

2.0 Introduction

The chapter provides background to the emergence and spread of the Covid-19 pandemic in order to demonstrate why vaccines were necessary as one of the strategies adopted to fight the pandemic locally and globally. The background will also serve to illustrate the rise and spread of Covid-19 denialists and sceptics who are blamed for the social media spread of anti-vaccination information. Both locally and globally misinformation and disinformation about the coronavirus disease, let alone vaccination, made use of digital media to campaign against what they considered a scientifically fictitious and non-existent disease. To them vaccination was not only a waste of resources in term of manufacturing vaccines but also an exercise fraught with danger to human health since the vaccine manufacture was rushed when the vaccines' long term effects have not been established. Against this background, anti-Covid-19 and vaccination was a strong lobby digitally and physically as seen through anti-Covid-19 restrictions throughout the world. However, informed by WHO's Covid-19 guidelines and protocols the government of Zimbabwe chose to fight Covid-19 misinformation and disinformation through especially media campaigns encouraging citizens to practice social distancing, masking-up, testing and above all to get vaccinated.

2.1 The Rise and Spread of the Covid-19 Pandemic

The coronavirus pandemic was first discovered in China in November 2019 in the town of Wuhan, Hubei Province. Then it was thought to be a localized disease or epidemic. Covid-19 was first detected in China. World Health Organization (WHO) China country office testified a collection of pneumonia cases in the city of Wuhan on 31 December 2019, Hubei Province of China, then 7 January 2020, causative pathogen identified as a novel coronavirus (2019-nCoV).

The new virus belonged to the corona virus family which consists of crown shaped viruses, hence the name 'corona' which means 'crown.' Owing to its outbreak in the year 2019 the disease was officially named Covid-19. Primarily transmission from one person to another was not apparent and the majority of the cases were epidemiologically connected to seafood and live wildlife market in Jiangnan District of Hubei Province. When cases in China began to decline, there was a rapid increase in other parts of the world like Europe and Iran, similarly cases in Africa began to rise as well. Although some countries were hit much harder than others, the entire world has been impacted (Finley, 2020:5).

Medically the corona virus disease was explained as being caused by a new severe respiratory syndrome corona virus 2 (SARS-CoV-2). SARS-CoV-2 is described as a betacoronavirus, which means it originated in bats. The envelope of the coronaviruses is enclosed with club-shaped glycoproteins which appear like 'crowns', or 'halos' – hence the name 'coronavirus.' (Andersen et al, 2020) Coronaviruses are responsible for the common cold, and usually cause self-limited upper respiratory tract infections. Thereafter, the covid-19 pandemic has continued to generate undesigned problems in humanity's different life issues. For instance, the politics, economies, spiritualities and social affairs of nations were shaken as government adopted WHO protocols and guidelines such as lockdowns, mandatory masking-up and social distancing, Covid-19 passes, and encouraged citizens to be vaccinated. Lockdowns had a huge toll on the economy with many companies closing down particularly in tourism and entertainment industries in which huge crowds normally gather and are blamed for being superspreaders of the corona virus. Religious gatherings were also outlawed and later extended only to the vaccinated while funerals and the rites they involve were banned and limited to just a few mourners. Owing to the reduced economic production coupled with the laying off of workers, many households experienced hunger and starvation. Meanwhile, the virus was spreading quickly and killing millions in especially western countries, the disease's psychological impact began to be felt. Fear gripped citizens and the coming of vaccines nearly a year after the outbreak of the coronavirus was welcomed by many citizens and their governments, although as we will see some citizens refused vaccinations.

In reality the world was plunged into a health confusion because medically no government's health system was prepared for the pandemic the nature of whose virus they scientifically medical experts did not know.

Although in 2022 the spread of the disease began to subside, many cases were recorded in China's economic capital, Shanghai, the United States, Bangladesh, South Africa, the UK, among many other countries. As of now official statistics show that 560, 781, 949 were infected worldwide while 6, 373, 141 succumbed to the disease. In Zimbabwe as of July 12 2022, 255, 981 people had been affected and 5 565 dying from the pandemic. Sadly these statistics include those of frontline workers such as doctors and nurses who ironically are supposed to be helping society to combat the disease.

Populations most at risk were later to be discovered to be the elderly and, people with underlying medical issues, such as heart problems or respiratory problems, HIV/AIDS, cancer, and other ailments. According to the most recent global numbers, people over 80 years old (14.8%), infected with the virus and have died from it, compared with people aged 40-49 (0.4%) in and none in children under nine years. As the disease spread, the situations across countries were rapidly changing with and these numbers cases of affected and infected people continuing to change. This reality came against the mutation of the virus. In mid-April 2021 the virus mutated into the delta variant first discovered in India and then spread to other parts of the world such as United Kingdom (UK). Currently the omicron variant which emerged in South Africa on the 24th of November 2021 is still dominant although epidemiologists believe it is the virus' last variant. It is less lethal as compared to the delta variant and does not require hospitalisation.

Medically the virus is spread from one person to another through small respiratory droplets. When a person coughs or sneezes, these droplets can reach the close surfaces. There is also indication that the COVID-19 can last on surfaces especially plastic or metal for up to 3 or more days. This is why people are advised to avoid contracting the virus focuses on hand-washing with soap, the use of sanitizers and keeping a distance from people who are infected. While the majority are seen wearing masks, especially in public, at present (WHO) instructs that one only

needs to wear a mask if they are not feeling well or looking after someone who is sick in addition to the imperative measures above.

2.3 The Global Vaccination Drive

The scale with which people were being infected by and dying from the virus, especially in western and Asian countries prompted the manufacture of vaccines in the shortest possible time. Effective vaccines normally take years to manufacture because they need to be tested to a selected population and monitoring their efficacy starting with a small number of people and then increasing the numbers. The side effects of the vaccines also need to be identified and see how harmful they are and also how they can be minimized. However, the rate at which people were being infected prompted pharmaceutical companies to fast-track the manufacture of vaccines. The first coronavirus vaccine was produced by Russia in 2020 and its efficacy being put at 92%. The Oxford-AstraZeneca vaccine was the most widely used worldwide.

However, the rapid manufacture of the vaccines and some of the side effects encountered such as blood clotting and death led some countries to suspend the vaccines. For instance, in Sweden, Denmark, Norway, Iceland, Bulgaria, The Democratic Republic of Congo, Ireland among other eleven (11) to make 18 countries, the AstraZeneca's vaccine was suspended. In South Africa vaccines also raised problems as they were associated with the death of patients (Jong, 2021:1). Besides those who were denying the existence of Covid-19 and regarding it as just another common cold, these isolated problems associated with vaccines intensified coronavirus denialists and sceptics' anti-vaccine drive. They did not only refuse to be vaccinated, they were mobilized citizens to demonstrate physically against vaccination and other strategies for combating the pandemic.



With digital media at their disposal, they also used the internet to rally support against vaccination posting purported negative effects of being vaccinated such as developing boils all over the body resulting in skin damage and disease (see picture above), death and other purported challenges.

2.4 Covid-19 and Vaccination Campaigns in Zimbabwe

In Zimbabwe Covid-19 was first detected in March 2020). Like other governments the Zimbabwean health system, which unfortunately was already facing insurmountable challenges such as strikes by medical staff, lack of drug supply and medical equipment, was thrown into a quandary by the coronavirus pandemic. Like other countries there were no adequate ventilators seriously needed to improve the breathing of patients let alone scientifically known drugs that could remedy the viral disease. By the time vaccines were introduced in February 2021 35,423 people had been infected.

China donated the first consigned of vaccines to Zimbabwe. The first 200, 000 doses of Sinopharm BIBP vaccine reached Harare's Robert Mugabe International Airport on the 14th day of February 2021. Thereafter, Zimbabwe launched its national Covid-19 vaccination program using Sinopharm BIBP vaccine on 22 February 2021. As of 12 July 2022, 6, 322, 448 people have received their first dose, 4, 683, 952 have received their second dose, and 908, 444 have received a third dose. The overall outcome being 42.5% of the Zimbabwean population received at least one dose and 31.2% of the Zimbabwean population fully vaccinated.

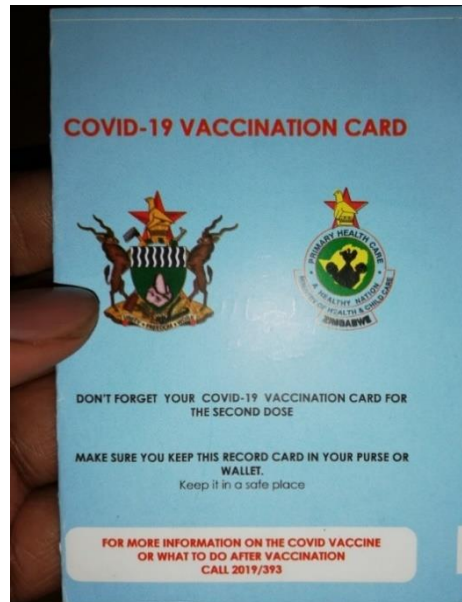
However, the low turnout has been linked to distortions circulating on basic social media platforms, thereby triggering vaccine hesitancy. Some people were not in a hurry to get vaccinated until they fell sick with Covid-19. People believed in misinformation and conspiracy theories about vaccines. For instance, there is (was) a roamer that when you get vaccinated you will die earlier than expected, other said by the time you get vaccinated you will be initiated into Satanism, most people are afraid of Satanism. These conspiracy theories are dangerous to a certain extent. Social media is a major weapon that recommends messages promoting the conspiracy theory.

Conspiracy theories are grounded on a diversity of rational patterns that are unreliable implements for pursuing the truth. Typically, conspiracy theories are not sustained by proof that withstands scrutiny but this does not stop them from developing. (Lewandowsky & Cook, 2020:3)

Conspiracy theories ruined the vaccination uptake in a variety of ways. For illustration, exposure to conspiracy theories have a negative effect on public's intentions to get vaccinated. Conspiratorial thinking is characterised by hyper doubting of all the information that does not suit the theory, over-interpreting proof that supports a desired theory, and contradiction. Social media amplifies conspiracy thinking. Allcott and Gentzkow (2017:211) state that, "Social media created a world in which anybody can potentially reach many people just like conventional media." Basically, conspiracy theorists are more disposed to like and share conspiracist information on Facebook, WhatsApp, and Twitter among other social media platforms. COVID-19 anti-vaccination messages are moderately prevalent in Zimbabwe and social media being a major threat in amplifying them hence, vaccine hesitancy.

Zimbabwe missed its target of vaccinating 60% of the 15 million population. According to a Health Worker interviewee, "The problem is in getting vaccination deliveries and convincing more people to accept them (vaccines)." To raise the expected total, the government made a compulsory for all its workers to get vaccinated. This involved harsh rules implemented by the Zimbabwean government as a way to foster and upkeep vaccination campaign. Those who were

not vaccinated were exempted from going into bars, places of worship, restaurants to mention but a few, until when fully inoculated. Those who were vaccinated were given a Covid-19 vaccination card (image below) to use as a permit when travelling.



However, in spite of the Zimbabwean government announcing a number of measures to limit the spread of the Covid-19 such as the no job no pay strategy not all workers were vaccinated because the majority are in the informal sector. The informal sector of the economy meant that many people in the working class category were not vaccinated. Then comes the rural population where again many people were not vaccinated owing to a number of factors that include religious beliefs and access to health facilities with the relevant vaccines. The fact that by July 2022, 6, 322, 448 people had been vaccinated out of a population of 15, 298, 328, it suggests that 8, 975, 880 were not vaccinated. Vaccine hesitancy was therefore high and it is the purpose of the next chapters to explain what contributed to this health challenge as far as the desire to vaccinate the Zimbabwean population is concerned.

2.5 Conclusion

The chapter was concerned with analyzing the emergence of the Covid-19 pandemic, its spread from China to other parts of the world including Africa and Zimbabwe, its devastation

capabilities and the necessity to develop vaccines. Like in all other countries, in Zimbabwe vaccination was adopted as a strategy to limit transmission, reduce death and develop herd immunity. The government set targets of inoculating 60% of the total population. However, by July 2022, 6, 322, 448 Zimbabweans had been vaccinated leaving an estimated population of 8, 975, 880 unvaccinated. Whereas the government was busy with its vaccination campaigns, social media was awash with anti-vaccination information which was embraced by pre-existing anti-western medicine believes in Zimbabwe and the rest of Sub-Saharan Africa thus greatly contributing to vaccine hesitancy. It is thus the intention of the next chapters to analyse social media information concerning vaccination and how it contributed to strengthening anti-vaccination conspiracy theories and pre-existing information against inoculation.

CHAPTER THREE

DIGITAL STRATEGIES AGAINST THE VACCINATION CAMPAIGNS IN ZIMBABWE

3.0 Introduction

In this chapter we are concerned with analyzing the various strategies adopted by both anti-and pro-vaccination campaigners in order to establish the messages of the two antagonistic sides' impact on vaccination campaigns in Zimbabwe. Various communication strategies that exploited digital technology's multimodal techniques that combine the textual, pictorial/visual, sound and the performative to disseminate information employed by social media users to spread anti-vaccination pro-vaccination information will be analysed in search of the effectiveness of the strategies in swaying people in favour of the information conveyed. The chapter also analyses pre-existing anti-vaccination information from religious groups that have anti-western medicine doctrines as one of the factors the digital campaigns against vaccination found anchorage.

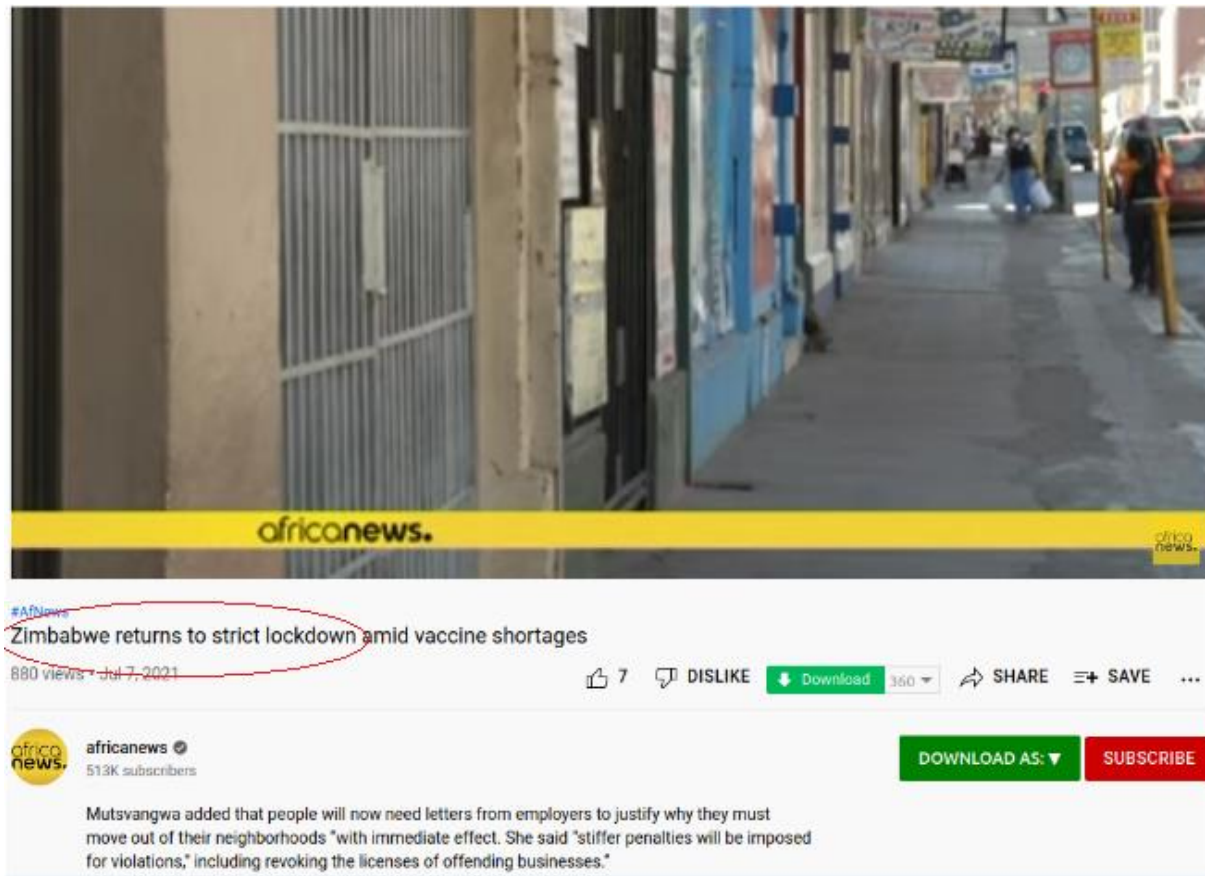
3.1 Social Media and Covid-19 Communication

This section is going to explicate the role that was played by digital products during the beginning of Zimbabwean lockdown in March 2020 up-to date. The 2019 coronavirus epidemic has received close attention from different people in Zimbabwe. During lockdown thousands of Zimbabweans were keen to acquire information regarding Covid-19 through digital texts. The speedy broadcasting of information related to Covid-19 has affected the behavior of the public during the pandemic. During the Covid-19 times, the digital texts play a central part in disseminating information as people were not allowed to meet physically. After physical interaction was banned in Zimbabwe people began to advance to more sophisticated modes of communication using digital tools. Major digital instrument adopted for communication in Zimbabwe was social media.

The so-called social media is part and parcel of web 2.0. Social media transforms people from content consumers into content producers, to be in precise, anyone can send anything anytime. Social media tools as digital text is generally accessible to anybody at slight charges, they are gadgets that permit anyone (including people from lower class) to publish or access information. From the time when O'Reilly applied the dictum web 2.0 in 2005 in a particular conference, people began to pay attention to it in several aspects of society, governments, and personalities. Curran et al (2012) astutely note that;

First notions on social media followed the cyber optimism that characterised the studies about the Internet around 1990. A number of studies highlighted the power of the digital tools and its capability to circumvent physical, socio-economic, political, and personal impediments.

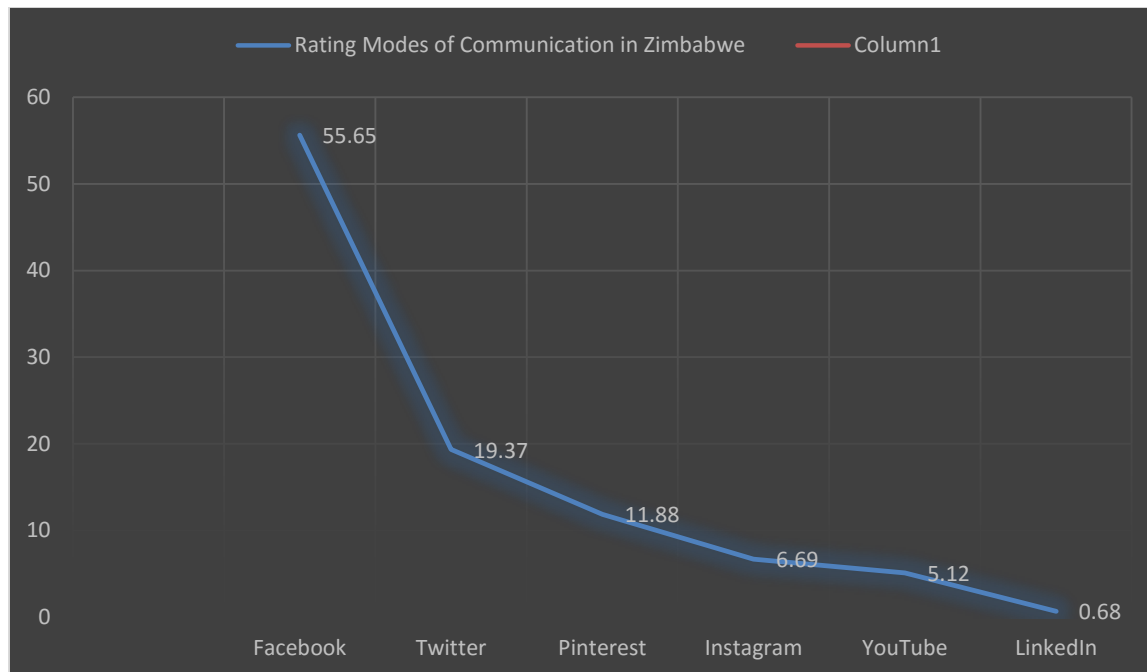
Social media was utilised to obtain information on Covid-19 virus more as compared to conventional media like television, radio, and newspapers.



As mentioned earlier, people were suspended from their physical or outdoor interactions and activities. Drawing evidence from a statement highlighted in the screenshot above posted and attached as a headline by Africa News on a YouTube video (<https://www.youtube.com/watch?v=cMaesvDDv-Y>), ‘Zimbabwe returns to strict lockdown...’ Evident from the aforementioned dictum is the deployment of the diction ‘strict’ by the YouTuber to reveal the degree of the harshness of lockdown. Lockdown is a state of isolation and restricted access or travel, for example, to contain an infectious disease or prevent movement of dangerous people. So during national lockdown people began to utilise the digital tools extensively since physical interaction was banned, candidly, people began to live life digitally. Digital products or digital goods are anything that one can produce or retrieve digitally. Digital goods are disembodied goods in the sense that they can exist without embodiment in a physical form. Any product one can produce online that does not have physical form or substance. Digital goods or e-goods are intangible goods that exist in digital form. Major digital goods that were utilised during lockdown were social media websites, music, videos, images, mobile social

media applications, and e-books. Communication technologies were very important tools for social interaction as they influence collective action during lockdown.

The changes to system of communication induced by the Covid-19 pandemic came amidst a time of considerable evolution in how people compose and send their messages, and how the receiver discover, evaluate, and select the exact information he or she wants. Almost for a decade, the use of social media and digital products remained neutral in Zimbabwe until Covid-19 struck where people began to utilise digital tools more widely and effectively. Many government and private institutions such as schools, colleges, universities, churches, and industries were closed. Some began to conduct their work online than ever before. In this case, social media is (was) much closer to people than conventional media. Social media is the internet and mobile technologies for interactive social networking. According to Wong et al (2021:1), “By definition, social media is ‘interactive computer technologies that enable the creation and sharing of information, ideas and other forms of expression via virtual and networks’”. As a communication machinery, the social media is used in vaccination campaigns to disseminate messages and contribute to update and involve citizens in the campaign. It is also on social media where various people began to create and spread messages regarding Covid-19 vaccines, not all messages were favourable, some information was against the vaccine and it instilled fear among people thereby triggering vaccine hesitancy. Cyberspace is a replica of the real world. Cyberspace support people’s need for communal contact, using internet technologies to transform traditional media discourses (one to many) into social media dialogues (many to many). According to Patrut and Patrut (2014:v); “the active members become leaders of opinion and inventors of conversations, that is, of word of mouth.” Technology is an instrument generated by man to help him live in community. Social media has energised and strengthened people by allowing them to come together by providing ways of communication and equipment for rapid exchange of opinions during the country’s (Zimbabwe) lockdown. The line below demonstrates the major ways of communication implemented in Zimbabwe. The results from the line are mainly directed to the urbanites both upper and lower class citizens.



Combining the communication channel from the line above, according to Social Media Stats Zimbabwe between June 2021 to June 2022, the majority of citizens used Facebook (55.65%) for receiving and sending information, followed by Twitter (19.37%), Pinterest (11.88%), Instagram (6.69%), YouTube (5.12%), and the least way of communication being LinkedIn with a very shallow percentage of (0.68%). Facebook have got the highest number of users in Zimbabwe because it is too interactive as compared to the other.

However, instead of utilizing the internet, to be in particular, social media for the benefit of the society people began to misuse it. In this case, during lockdown some people began to spread false information and news regarding the Covid-19 pandemic *per se*. Not all people abuse the platform, other utilised the social media platform effectively. There are different reasons why people send different messages in whatever manner they do on social media. Users that create content on social media are inspired by issues that comprise peer interrelation, the chase of popularity, notoriety, and most significantly self-establishment. This is a major reason why governments and organisations (for instance, the European Union and the Digital Services Act) are creating legislation for responsible use of social media. The Minister of information Monica Mutsvangwa has urged Zimbabweans to use the cyber-space legitimately if they want to remain relevant in the 21st century world. She said this while speaking at the Marketers Association of

Zimbabwe's 13th convention took place in Kariba which ran below the topic, *Marketing in the Age of Artificial Intelligence*. Social media has been utilised by some people to spread propaganda, disinformation, hearsay, and conspiracy theories about Covid-19 vaccines. (Newsday, 2021)

3.2 Pre-existing Anti-vaccination Attitudes in Zimbabwe and Africa

Vaccines are controversial drugs globally, particularly to cultures and populations of strong religious beliefs which consider faith healing as more preferable than medical therapy. Besides that, in the so-called western scientific cultures, there are many populations that question the efficacy of vaccines and consider them as harmful to people's health. In Zimbabwe and Africa where western medicine is usually difficult to access due to poverty (people have to pay to access diagnosis and drugs at health facilities), many people rely on traditional medicine as it is offered by herbalists and traditional healers. When Covid-19 arose many local remedies were resorted to including the use of garlic, ginger, lemon, onion and zumbani (See picture below). In 2020 the government of Madagascar produced the first herbal based local remedy to Covid-19 called Covid-Organics while in other African countries like Zimbabwe many local plants were sold at the markets as remedies to the new virus.



Open Parly ZW on Twitter:
"Commenting on traditional herbs



Zimbabweans pin hopes on woody
shrub to beat COVID-19

Zumbani as a media spectacle

Believers in traditional medicine were one of the sections of the population against vaccination. In the DRC for instance, Covid-19 vaccines expired because they had no takers as people did not prefer to be vaccinated.

Independent African Churches were another section of the population that is against the use of western medicine or the seeking of health solutions at modern health facilities. The Johane Marange Church is perhaps the most well known for its stance against seeking treatment at western medicine based health facilities. They staunchly believed in the verse when Jesus was preaching in Galilee, teaching people in their synagogues informing them that God is able to heal all kinds of sickness and all kinds of diseases among the people (Matthew 4:23). Many others only visit health facilities as the last resort when faith healing has failed to cure a patient. In the Marange sect pregnant women never attend antenatal clinics but deliver under the care of a church midwife. Followers of the sect and others are notorious for refusing to have their children vaccinated as approved by WHO and the government. Vaccination teams have been harassed several times for approaching homes of such believers.

Among the various explanations behind vaccine hesitancy, religious issue is one of the vital reasons specifically, the Apostolic sector. According to the Ministry of Health and Child Care (2017:39); the apostolic canon on non-use of contemporary healthcare facilities and medications (comprising vaccines) affects healthcare seeking individuals and tends to emphasize religious and traditional ways of healing such as faith healing rituals (prayer, holy water, apostolic traditional birth attendants etc.) This issue gave rise to cyber-critics who participate in different social media platforms commenting the issue of abstaining from using medical facilities. A screenshot below clearly divulges one of the WhatsApp group member with the username Sir Jefu, after a poster regarding the death of the church leader Taguta in mid-April 2022, he began to blame Johanne Marange members to cause severe deaths dodging vital health facilities.

+263 77

~Sir Jefu

State assisted funeral for
Johanne Marange high priest
Taguta...



But how Taguta may be tagged a spiritual father to the nation whilst his church had bad issues pertaining to the disrespecting countries laws due to its child-marriages and high mortality rates as a result of voiding medical issues???

7:34 am

Despite the fact that Covid-19 vaccination programme was an exercise being conducted by the Zimbabwean government, the Marange sect persists against vaccine. According to the Vice President of Zimbabwe and Minister of Health, Constantino Chiwenga, “Zimbabweans get vaccinated to protect themselves against the virus that took lives of 5 457 citizens and 6,18 million worldwide” (The Herald, 10 April 2022). To be in particular, the Johanne Marange Apostolic Church shun the usage of modern healthcare services. In these churches death is the will and law of God. According to an interviewee, “In our religion, people are not allowed to go to the clinic. If a person dies, God will give you another one.” From the indicated statement, the religious doctrine on non-use of modern healthcare services including Covid-19 vaccination increases the risks of apostolic personnel to preventable illnesses and deaths. Kant et al (2021:55) notes that, ‘Some religious leaders spreads conspiracy theories and misinformation about vaccines, claimed the vaccines are unusual and thus irreligious.’ For instance, there is a post on YouTube regarding Covid-19 vaccine, a person with the username Seagull Club was inspiring other people that even the diseases are controlled by God and he even quoted the Bible (Matthew 4:23-24).



Seagull Club 4 months ago

In Jehovah God's new world in the hands of Christ Jesus all sicknesses, infirmities, disabilities, and even death will be eliminated!! 🙏🙏 Matt 4:23,24 Jesus cured EVERYONE

From the aforecaptured snapshot, other people have beliefs and they are not scared of death. When a person dies they say it was God's will and there is nothing that overpowers God. Drawing from the screenshot above, the application of the dictum '...all sicknesses...' encompasses the coronavirus diseases. These people believed that their faith would be sufficient to withstand prevent contracting diseases and also have their well-being restored after they are infected.

However, not all apostolic sects denounces Covid-19 vaccine, others urged their congregants to be immunized. Other church leaders after knowing the benefits of vaccination they began to strengthen and motivating their followers and encouraging them to get children vaccinated. One of the leading apostolic church namely The African Apostolic Church urged its congregants to get vaccinated. This was captured on one of the followers' Facebook group (DZANETSA MUCHECHI AAC [Critical Issues in Church]), a post traceable on the link <https://www.facebook.com/search/top/?q=vakabaiwa> , was sent by a group participant on the 13th day of August 2021 saying in the Shona vernacular, “*Vakabaiwa Vaccine vese tine gumano mangwana kuChashawasha grounds, vasina munoona yekutamba. Ndatenda!*” (To all those who have been vaccinated we have a gathering tomorrow at Chishawasha grounds, the unvaccinated you will see what you can do. Thank you!) This is imitated by the screenshot below;

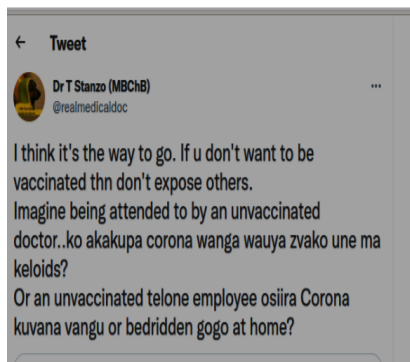


From the above statement, it is clear that, other churches specifically the apostolic sects urged the congregants to be vaccinated. By the application of the third person plural collective voice (va-), *Vakabaiwa vaccine vese...* it is simply a voice that is appealing to a large number of congregants. However, since some of them are not the dominant ones such as Marange, Chishanu and others, it meant the religious anti-vaccination message remained strong. Therefore,

any information or conspiracies against vaccination find fertile ground among followers of such independent churches. There was therefore vaccine hesitancy and resistance before the coming of Covid-19 arising from the historical anti-western medicine tradition among both believers in African medicine and faith-based therapies.

Besides religious and traditional factors promoting vaccine hesitancy there are also socio-economic factors. In spite of the religious reasons to Covid-19 vaccine hesitancy, this research also clarifies that, vaccine hesitancy was (is) amongst the less-educated, females, ethnic minorities, people living in rural areas, and those with lower income or economic security. A number of the less privileged people were indeterminate about the effectiveness of the vaccine and lacked self-assurance on the efficacy of the vaccine. Most of the underprivileged class in Zimbabwe seems to work in the public places which are particularly vulnerable to the infection risk of a virus contagion or contamination.

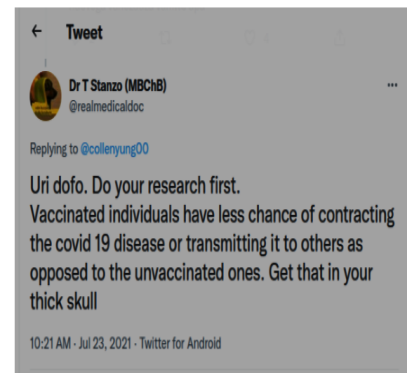
1



2



3



Education level is another factor linked to hesitancy in Covid-19 vaccine uptake in this research. Marzo et al (2022:6) notes that, 'Members with inferior education were notably more likely to express hesitancy in getting Covid-19 vaccines than members with tertiary education'. This is undoubtedly revealed by the three screenshots above where there was a misunderstanding on Twitter regarding covid-19 vaccine and both sides (for and against vaccine) were accusing each other of being uneducated. The participant on picture two (2) with the name Co Mak was

interrogating the former on picture one (1) with the name Dr. T. Stanzo (MBChB) regarding the effectiveness of the vaccine when he said, ‘...*inga vakabaiwa vanongobatwa nayo and vanozaadza vamwe...*’ (...even those who are vaccinated they can still be infected and infect others...), alternatively on picture three (3) he was accused of being dumb and uneducated in a statement that reads, ‘*Uri dofo* (You’re dumb). Do your research first. Vaccinated individuals have less chance of contracting the covid 19 diseases or transmitting it to others as opposed to the unvaccinated ones. Get that in your thick skull.’ By the term and a mocking expression *dofo* (dumb) and ‘get that in your thick skull’ respectively, Co Mak in this case represented those who were not willing to get vaccinated was referred to as a dull person with a gloomy state of mind and ill humor. Additionally, the odds of intending to get vaccinated ‘...had high level of education, and identified vaccination and face mask usage as stoppage measures.’ (McAbee, Tapera & Kanyangarara, 2021:1) In Zimbabwe during the Covid-19 pandemic, persons with higher levels of education have tended to get vaccinated.

This study also anticipated that there is an inverse relationship between poverty and vaccination uptake, besides educational factor. People who are poor are more probable to have poor health circumstances and often have deficient health care. Some people especially from the lower class highlighted that they face(d) challenges in accessing the health facility because of long distance to health facilities. Long distances to the health facility require the usage of transport, which is costly for several poor people. The distance to health facilities was one of the barriers to uptake of covid-19 vaccine. Poverty rates are much higher in high density suburbs. According to an interviewee, ‘*Pakatanga zvekubaiwa izvi inzvimbo shoma dzaibairwa vanhu uye dzaive kure saka waifanira kutokwira kombi sezvo ruzhinji rwedu rwusina motokari*’ (When the vaccination program began there were few vaccination centres again they were a distance afar so we have to board a bus because most of us do not have private cars). Again, some people have a lot of children of the same ages and they do not have someone close to leave these children with. The hospital is far away that they cannot carry all the children to the vaccination centre. There was a mythical statement that circulated on social media that says in the Shona vernacular, ‘*Vanhu vane mari ndivo vanobatwa necovid*’ (Rich people are the ones who can be contaminated by the covid virus). This is obvious that this myth was created by a person from the lower class.

Moreover, major reasons for low vaccine uptake in high poverty communities include language barriers and a lack of trust in effectiveness of the vaccine.

3.3 Anti- versus Pro-vaccination

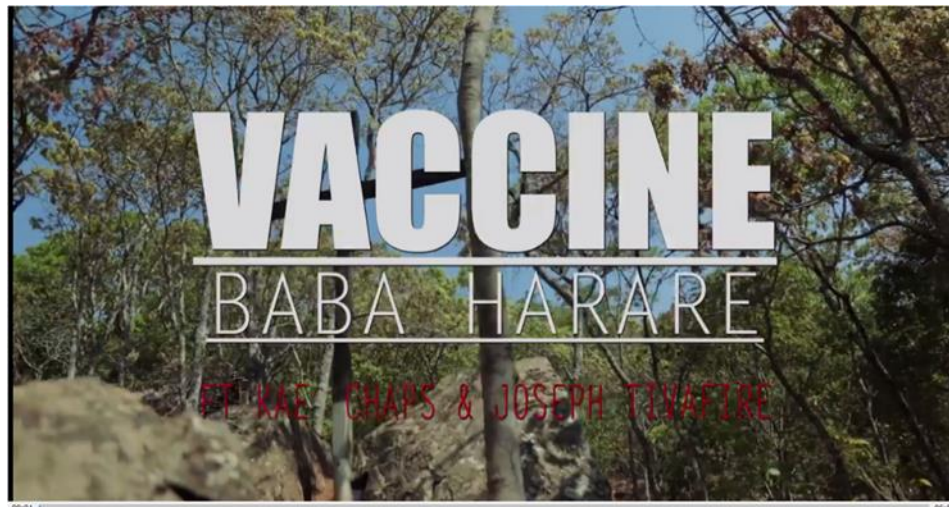
In this section we comparatively analyse two categories of social media messages, one that was anti-vaccination and thus promoting vaccine hesitancy and the other that was pro-vaccination. The intention is to demonstrate that in the fight against Covid-19 there was a propaganda war that exploited social media platforms to launch its arsenals. The two camps resulted in two frames of meaning that were witnessed on different social media platforms during the ravaging Covid-19 pandemic. Just like in other countries, in Zimbabwe the cyberspace became a fighting domain between those people who were against the vaccine and those who were willing to get vaccinated. Besides social media platforms, right from the beginning of the national lockdown in Zimbabwe, people received emergency alert text messages transmitted to their cell phones especially by Econet Wireless the country's leading telecommunication company. The message, known as SMS (Short Message Service), an unblockable message system, urged and informed people about the Covid-19 daily updates and major preventive ways to the virus. However, it appears the anti-vaccination camp employed linguistic and cultural arsenals more effectively than the pro-vaccination camp thus contributing to inoculation hesitancy.

3.3.1 Pro-vaccination messages

In order to minimise the spread of Covid-19 it was found important to develop a communication strategies to counteract “vaccine hesitancy”, that is, the deferral in receiving, or denial of vaccination in spite of the availability of vaccination services. Media framing was a major mode of communication that was adopted by people as a way of spreading their messages regarding vaccine perceptions. A ‘media frame’ refers to “words, pictures and presentation styles that a presenter (for example, a politician, a media outlet) uses when transmitting information about an issue or incident to listeners” (Chong and Druckman 2007, 100). Therefore those people who received information about the efficacy of the vaccine were more probable to report that they were going to be vaccinated, those who received information that others were uncertain to take

the vaccine were more likely to report that they themselves were not going to get vaccinated. Communication of positive effective messages about the vaccine was deemed important for public health organisations that seek to promote vaccine uptake in Zimbabwe in particular and the whole world in general. French et al. (2020) state that the Covid-19 pandemic, as in forthcoming occurrences of vaccine-preventable illness, it will be vital to avoid this reluctance and to endorse vaccine uptake using operative communication approaches.

Understanding diverse kinds of information may manipulate the public's opinions and vaccination objectives are required in order to develop an effectual communication plan. Communication about the advancement of Covid-19 vaccination program in Zimbabwe was transmitted through "frames" used in the messages on basic social media platforms. Pro-vaccination messages made use of music in the form of framed videos that were posted on YouTube both of positive and negative frames. YouTube began to autoplay several videos urging people to get vaccinated. Art (visual art) became a very convenient weapon and instrument in disseminating news and information concerning the effectiveness of the vaccine, thus the functionality of theories of visuality. Such visual subjects might be the participants in a performance in a video and usually taken from their own point of views. Music-videos took centre stage in disseminating information on vaccines. A song is fruitful when it turns out to be video, the alignment of music and motion pictures. Chiwome (1996:2) notes that, "Songs are the highest form of poetry." This statement is clearly depicted by the song, "Vaccine" (2021) by Baba Harare featuring Kae Chaps and Joseph Tivafire traceable on this link (<https://youtu.be/bLYZ2StecP4>) on YouTube. Due to the usage of vibrant poetic devices and rhetoric language by the persona, he (Baba Harare) revealed the usefulness of the vaccine by indirectly caricaturing it to a religious prospect.



Verse two: 02:49 of 06:57

Lord my spirit sanitize

Vavengi vangu isai mulockdown

Kana varoi vese muquarantine

Ini mondijunga zivaccine

Vaccine yomweya mutsvene

Chorus

Mwari ndibaye iwo vaccine

.....

.....

Ndoda kujungwawo vaccine

Ndoda kujungwawo vaccine

The artist is asking strength from God to conquer all his enemies (virus). Going back to the Old Testament this is the same strength that was given to Moses at the burning bush and Isaiah in the temple (Exodus 3 vs 12 and Isaiah 6 vs 6 respectively). There are two words that are applied by the artist characterised by antagonism like *Vavengi* (enemies), *varoi* (witches) and they are a replica of the covid virus. Still in the same vein, the persona uses repetition of the expression '*Ndoda kujungwawo vaccine*' (I want to be vaccinated)' in the chorus and parallelism. The repeated use of the same word or word pattern as a rhetorical device by the artist simply mean he is sympathizing the effectiveness of the vaccine. In this case the covid-19 vaccine is being

framed in a positive manner. This song significantly promotes coronavirus disease 2019 (Covid-19) vaccines. As individuals increasingly turn to social media for health information (Lee & Jin, 2019), both national and local agencies have maintained an active voice on social media to encourage the uptake of Covid-19 vaccines. An artist is the mirror of the society. Literature do not grow in a vacuity but it is given shape by certain social and economic activities in a particular community.

3.3.2 Anti-Vaccination Framing

Messages against vaccination used the same digital techniques. For example, the screenshot/image captured on Facebook below subsidizes to the understanding of the deleterious influence of message framing on vaccine uptake.



As noted above, a particular Facebook user with the name Linda Melisa posted an image on October 13, 2020 with a statement that says in the Shona vernacular, '*Nyangwe kunyashorwa*

hatibaiwe zvekudaro abaiwa achabuda ndonzi chihera mukonde porisi sungai henyu’ (Even if we are looked down upon we are not going to get vaccinated, the one who is vaccinated will quit, I am *Chihera*, police you can arrest me). This statement reflects refusal of the vaccination program and this is clearly revealed by the application of the third person plural collective voice (ti-), *hatibaiwe zvekudaro*, represents the voice of a large population. An excessive resistance to Covid-19 vaccines is also emphasized by the application of a cultural or totemic allusion ‘*Chihera*.’ They are known as *Vahera* (females) and in the current society are known to produce infidels, cheating wives and they challenge husbands and dominate their homes entirely. In Shona society women of this totemic epithet are considered a violent and at times can go against the established laws and norms. The name ‘*Chihera*’ itself means a lot regarding vaccine resistance and uptake. Kahari (1994: 21) states that ‘‘Man is name, name is man....as a measure of the importance I attach to names and characterisation, and to their ideological implications.’’ There is power in the spoken word. Nommoic creativity is a rhetorical discourse approach that focus of the generative and creative power of the word (Asante, 1998 and Ani, 1980). The generative and creative power of these names become the entry point for this study’s analytical engagement. The observed statement described and explicated above subsidizes to the understanding of the negative influence of message framing on vaccine resistance by common people. Framing theory elucidates that the media generate the selected frame by presenting information with predefined and constricted contextualisation. Therefore, the inspiration of a frame means to draw attention to certain aspects of the items of news coverage.

Covid-19 vaccine hesitant groups on social media have an startling footmark, with studies from 2020 to present-day showing that vast proportions of framed information about vaccines on prevalent social media platforms like Facebook, WhatsApp, Twitter to mention but a few, are anti-vaccination messages. Crafting frames for stories is frequently a mindful choice by writers, reporters, journalists or editors. Below is a tweet that was sent by Ignatius Mabasa in a common old epic story of the two main personified ‘*Tsuro naGudo*’ (Hare and Baboon).



Hare: Is it true that Hedgehog has Covid!

Baboon: Hedgehog is one of us, may he recover soon.

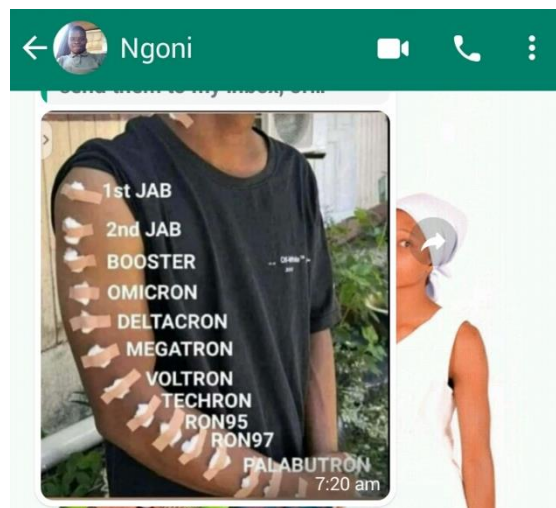
Hare: But did you know that he was not vaccinated?

Baboon: Why he did not get vaccinated but he was the one who vaccinated others?

Hare: You can show others way but you cannot use it.

The whole conversation of the two animals is interrogating vigorously the efficacy of the Covid-19 vaccine. People lacked trust in the government's ability to promise the effectiveness of the vaccine. There was a strong anti-vaccination movement in Zimbabwe that was witnessed on various framed messages on social media. This research explore that message frames play an immense role in vaccination attitudes and behavior among people. Price, Tewksbury and Powers, 1995:4 note that, 'The framing arrangement of occasions and information in the mass media can thus systematically influence how receivers of the information come to recognize these occasions'. Any social media message that contains forceful language(s) is more likely to increase psychological reactance, which may significantly weaken the effect of persuasive.

Just like on Twitter and Facebook, the issue of the effectiveness of Covid-19 vaccines has divided Zimbabweans on WhatsApp platform. As vaccination was (is) a preventive way, gain-framed information is thus more likely to be effective in increasing vaccine uptake. Not all media frames were circulated in form of words on WhatsApp, alternatively some information was in form of images that a sender uses when sharing information about a controversy. After previewing a certain image people will begin to draw their attitudes regarding vaccines from the set of provided beliefs stored in their memory. A good illustration is displayed on the screenshot below taken from a certain WhatsApp chat of an image that went viral of a person's right hand that has with about 14 vaccine jabs.

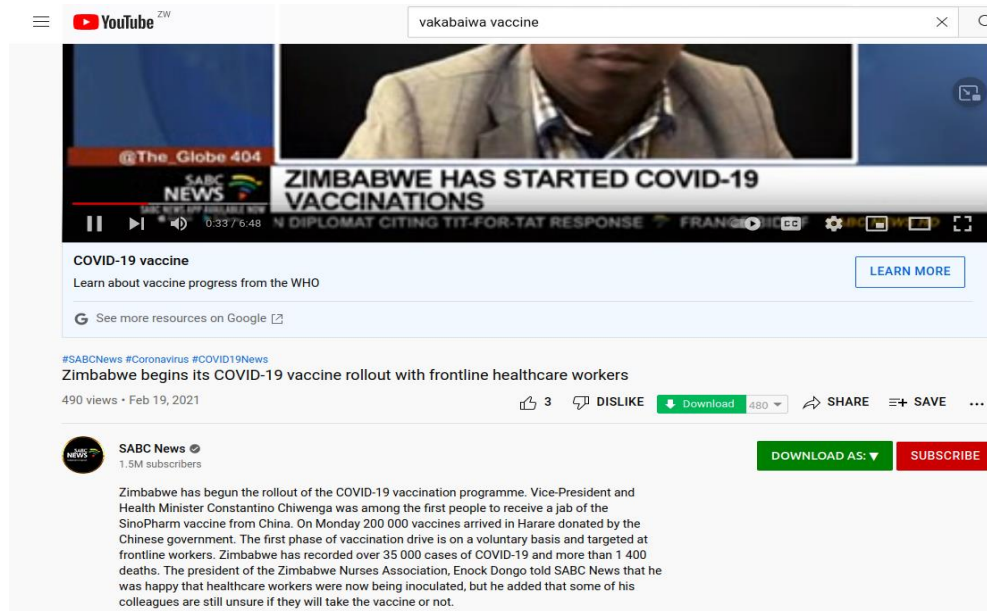


The above image instilled lack of trust on the efficacy of the vaccine among people. Having a lot of jabs on one hand it simply mean that one has to get vaccinated repeatedly, vaccination became an ineffective process thereby useless to people. The effectiveness of the vaccine is extremely important in the decision process, people interpret what is going on around their world through their principal framework below the topic of Frame Analysis (Goffman, 1974). Frames in the case of the above picture was designed to enhance understanding of the effectiveness of the vaccine.

3.4 Covid-19 Vaccine Spectacle and Vaccine hesitancy

This section, drawing upon the work of Guy Debord, interrogates the Covid-19 vaccination campaign's spectacular framing in the context of sophisticated media and its high audience impact. The objective is to establish the extent to which the media spectacle promoted vaccination or actually served to undermine its intended objective. Specifically this section analyses how Covid-19 vaccine is captured and mediated to a watching world. According to Frew (2014: 1), "In a time when the explosion of the new media phenomenon globally transforms and networks modern life, it is appropriate that events and festivity are placed under the powerful gaze of Debord and Zizek and allowed an alternative voice." Technology is now unified into the everyday and realism for many is one of mediated cultural co-created convergence. The issue of covid-19 vaccine was (is) constantly captured and distilled into a perpetual cycle of digital-communities or networks such as WhatsApp, Facebook, Twitter or YouTube. Covid-19 vaccine was seen becoming headlines of different social media platforms and it became a debatable topic among the users. This study briefly addresses the nature of debates surrounding the Covid-19 vaccine on social media.

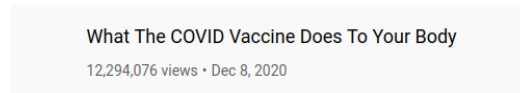
On YouTube, the ease of viewing and downloading videos on digital devices rapidly enhanced the YouTube audience, hence, contributing to the rise of media Covid-19 vaccine campaigns from different people in the cyberspace as well. There were various headings posted on YouTube regarding the vaccine and these statements impacted people in the process of vaccination uptake. YouTube videos are one of the greatest persuasive channel for different parties on vaccine to disseminate their agenda as it can be easily accessed by large audience. The picture below is a screenshot taken from YouTube channel of a video posted by SABC News on February 19 2021 with a heading that reads, Zimbabwe initiates its Covid-19 vaccine rollout with frontline healthcare workers'. The headline and video per se awakens people specifically Zimbabwean citizens on the orientation of the vaccination program.



In the case of the above picture, Covid-19 vaccine became a technologically mediated event, in which media form like YouTube process vaccine event in a spectacular form. The media is becomes more technically stunning and is playing a prevailing part in everyday life with cyberculture producing new sites like YouTube. Due to the number of its (SABC News) subscribers that counts up to 1.5 million, the heading that is attached to the video (<https://www.youtube.com/watch?v=fO78wh08uAQ>) in the screenshot above attracted a lot of attention among YouTubers. Frankly, the heading attached to the video is indirectly motivating citizens to get vaccinated, with the fact that a lot of people did not want to get vaccinated due to the lack of trust in the effectiveness of the vaccine, hence, the YouTuber willingly use healthcare workers as icon in the vaccination campaign to guarantee the effectiveness of the antidote. According to Kellner (2009:1);

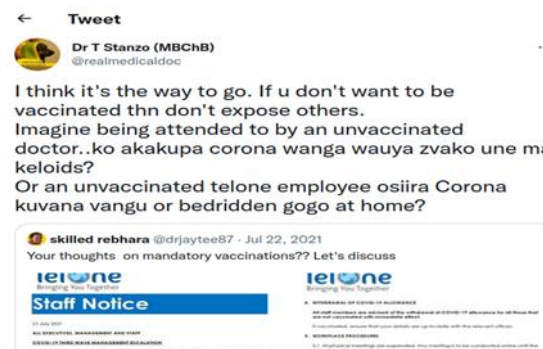
In time of intense competition with everyday cable television networks, radio, Internet sites, and ever proliferating social media platforms like Facebook, Twitter, YouTube, and Twitter, rivalry for consideration is ever more intense leading the corporate media to go to spectacularised certain issues, until the next spectacle emerges.

Covid-19 vaccine was increasingly mediated by media spectacle. Vaccine campaign and updates have exposed to the sense of spectacle in the epoch of media sensationalism. YouTube in this case, became one of the most powerful radicalizing instruments of the 21st century.



What keeps people glued to Covid-19 vaccine videos on YouTube was (is) its algorithm that seems to have concluded that people are drawn to vaccine content that is more extreme than what they started with. For instance, after searching the meaning of covid-19 then YouTube automatically drags you to a topic that says, 'What the Covid Vaccine Does to Your Body' (Picture Above).

As opposed to YouTube, Twitter is interesting in that many of mediated information are public and so people can get a view into the dynamics of dialogues and how they differ from other tweet behaviors. This study discovers that the vast majority of active conversations involve two or more people despite the public nature of these tweets. People were exposed to information about Covid-19 vaccination from different tweets on Twitter. In Zimbabwe, Twitter plays a significant role in facilitating the exchange of relevant information regarding the vaccine during Covid-19 pandemic. The widespread public adoption of Twitter as a tool for information seeking has led to debates about Covid-19 vaccine thereby playing an important role in vaccination decision-making. A picture below is a screenshot taken on Twitter sent by a participant with the name Dr T Stanzo (MBChB) regarding the disqualification of unvaccinated workers from TelOne. The heading of the notice reads, 'Covid-19 Third Wave Management Escalation'.



Staff Notice

21 July 2021

ALL EXECUTIVES, MANAGEMENT AND STAFF

COVID-19 THIRD WAVE MANAGEMENT ESCALATION

The company takes note of the intensification of the COVID-19 situation in the country and the consequent response by the Government in terms of decongestion and vaccination.

Building on this and our response plan that has been guiding our operations to ensure everyone's safety and business continuity, the company wishes to guide all management and staff as follows:

1. OFFICE DECONGESTION

As an essential service provider and to ensure that we do not compromise service provisioning to our clients, staff rotation will remain in place as per previous guidance. Under the arrangement, teams are divided into two groups with one group working from home and another working from office using one-week rotation cycles.

2. IMMEDIATE LEAVE FOR UNVACCINATED STAFF MEMBERS

Due to the reduction of business as a result of the reduced service uptake by both Enterprise clients and Government sector, the company is experiencing a decrease in the productive hours. As such, staff will be required to immediately take vacation leave starting with those that are **not vaccinated**. This is especially because unvaccinated individuals are at the highest risk of the effects of the COVID-19 virus, while also posing a great risk to all other staff and the company's business continuity.

Line management will be expected to provide an update to Corporate Services by 23 July 2021 in this regard.

3. VACCINATION

It is key for staff to ensure that they are vaccinated as this is a proven means of managing the impact of the COVID-19 virus. Staff are advised that besides this being good protection for themselves, the company may soon take further measures against the unvaccinated in order to protect vaccinated staff members and the business.

Evident from the above, during Covid-19 times certain debates were increasingly played out on the screens of media culture, which display spectacles like the Covid-19 vaccines. Soon after TelOne issued a notice on social media regarding the expulsion of the unvaccinated employees from work, the issue became a spectacle on different social media platforms, in this case Twitter. This is revealed on section number two (2) in the aforementioned TelOne notice in bold; **‘IMMEDIATE LEAVE FOR UNVACCINATED STAFF MEMBERS’**. This became a debatable event on social media, precisely, Twitter in this case became a war fighting domain between those who were against the vaccine and those supporting it. ‘Due to the development of new multimedia and information technologies, technospectacles have been definitively moulding

the forms and trajectories of contemporary societies and cultures...' (Kellner, 2005:58) Covid-19 vaccine issue was (is) increasingly mediated by media spectacle.

However, the reality the media spectacle regarding the coronavirus vaccination indicates is that the constant and continued spectacularisation of vaccination was reacting against disinformation and misinformation from anti-vaccination and Covid-19 denialists which had a huge following. The TelOne message posted on social media serves to indicate that the threats it communicates to workers indicates that many workers had shunned vaccination. The same applies to the no job no jab approach adopted by government for civil servants. It meant that many were not getting vaccinated in fear of the known short-term and the unknown long term effects of the vaccines. People are likely to accept adopting something when the message is delivered through persuasion rather than force and threats.

3.5 Social Media Impact on Vaccine Uptake: Applying Social Media Indicator (SMI) and the Mozdeh Software

In order to establish the impact of Social Media on vaccination, the researcher conducted social media empirical research during the research process. Firstly, the researcher was concerned if there were practical marks that Social Media usage by citizens in Zimbabwe has an outcome on herd immunity. By comparing Social Media engagement of people who were against the Covid-19 vaccine on one hand and those who supported it on the other hand, the researcher was able to calculate if the nature of their relationship; that is, which one had greater numbers than the other on selected social media platforms.

3.5.1 Social Media Indicator

The Social Media Indicator (SMI) is a useful instrument in illustrating the level of commitment of those people who were against the vaccine on social media and the extent in which they interact with other people on social media platforms, specifically the impact of their messages. The major indicating queries from the Social Media Indicator are presented in Table 1 below.

Table 1. Social Media Indicator

Social Media Indicator (SMI)
How many group participants are in a certain WhatsApp group?
How many replies of a Covid-19 vaccine message on WhatsApp?
On personal YouTube channel, what is the view count?
On personal Facebook account, how many friends?
Does the anti-vax campaigner have a profile at Twitter?
Based on 5 tweets, how many retweets?
In case of personal Twitter account, how many tweets?
Based on latest 10 tweets, how many replies?
How many following on Twitter account?
On Twitter account, how many followers?
In case of personal Facebook account, how many likes?
Does the vaccine campaigner have a YouTube channel?
How many videos are posted on this personal channel?
Based on all videos, how many times are they watched?
Based on all videos, how many comments?
Based on this channel, how many subscribers?

Because of the large number of different Social Media platforms, the researcher did not try to cover all social media in the analysis. It is more important to include the social media with the highest reach in Zimbabwe, for instance, WhatsApp, YouTube, Facebook and Twitter. WhatsApp is still the biggest social network site in Zimbabwe. By focusing on the measurement of contribution levels of the anti or pro-vaccine campaigners and calculating the interaction with other people, this research include all levels of contribution including e-enabling, e-engaging and e-empowerment.

By detecting the range of predefined standardised indicators, one can estimate a complete Social Media Indicator score for each anti- or pro-vaccine campaigner. This score is compared to the number of people who were vaccinated in Zimbabwe. The researcher calculated the SMI of the number of participants in a certain WhatsApp group where vaccine information was sent

(SMI=230/265). In active WhatsApp groups half of the participants would engage to a certain post. In case of organisation YouTube channel, there will be at least 500 views and 2, 000 subscribers. For instance, Baba Harare's song Vaccine have 551, 000 on the 14th of July 2022. In case of personal Facebook account, a person typically have at least 300 friends but if a person is popular in this case a musician or politician he or she have at least 2, 000 friends and followers. Most of the anti-vaccine campaigners have Twitter accounts and some of them use ghost accounts. Each tweet had at least 5 retweets. The SMI information provided above is only limited to Zimbabwean context. By observing the set of predefined standardised indicators, there are (were) more Covid-19 anti-vaccine campaigners as compared to pro-vaccine campaigners. Due to these results it is obvious that there was a rapid dissemination of anti-vaccine information leading to vaccine hesitancy.

3.5.2 Analysing Social Media Data Using Mozdeh Software: The Twitter Example

As a big data management tool, Mozdeh Software can be employed to analyse various posts with long texts extending over several pages. It is a convenient digital equipment for researchers to help easily differentiate the focal thoughts from tweets, or they simply highpoint the trending topics and the hyperlinks among them. According to Thelwall (2018:7), "Mozdeh gathers social web texts straight from websites that share them, such as Twitter and YouTube, or can easily import texts that have been collected in other ways." Twitter share its information through an Applications Programming Interface (API) which is a data allocation technology that Mozdeh cooperates with to download messages. The screenshot below reveals the search results for the query 'Covid-19 vaccines' for a set of Zimbabweans' tweets.

Mozdeh: covid-19 vaccines - 1343 matches out of 13239 (2.6%); vaccines for covid-19: 1.5160; av neg: 1.8484

File Analyse Networks Data Subprojects Advanced About

ID	Year	Text (abbreviated) Click Next Page ^ for more	Sentiment analysis results
			Score Pos Neg
000963(1)	2020	zim preparing for covid-19 vaccine rollout president	1 62.10% 45.77%
000982(1)	2020	who press release global equitable access to covid-19 vaccine estimated to generate economic benefit of at least bn in and bn by i	2 25.95% 30.81%
000987(1)	2020	covid-19 vaccine should be	3 10.50% 17.20%
002318(2)	2021	a consignment of dose of the covid-19 covaxin vaccine donated from india covid-19 is real get vaccinated and stay safe	4 1.17% 5.83%
002337(1)	2021	consignment of million dose of the covid-19 sinovac vaccine donated to the country by the people republic of china @china_amb_zim	5 0.29% 0.58%
002434(1)	2021	covid vaccine deployment and roll out plan the government has purchased on top of the sinopharm vaccine dose which were donated do	
002436(2)	2021	d n chiwenga receive the consignment of sinopharm vaccine dose donated to zimbabwe by the republic of china at the mugabe internat	
002438(1)	2021	at rgm international airport to receive the sinopharm vaccine donated by china to zimbabwe	
002442(1)	2021	visit for covid vaccine deployment and roll-out plan	
002453(1)	2021	reply @mohocczim meanwhile the vaccine has since been dispatched to all the country province where vaccination will begin on monday	
002472(1)	2021	the government of zimbabwe will roll out covid-19 vaccine for free to all our citizen who may wish to be vaccinated he spoke during	
002476(2)	2021	covid-19 vaccine update dr j c mangwiro deputy minister ministry of health and child care	
002487(1)	2021	gradual increase in covid-19 vaccination the covid-19 sinopharm vaccine rollout continue across the ten province the midland provi	
002488(1)	2021	cabinet approve four covid-19 vaccine	
002495(1)	2021	visit the @mohocczim website for information on vaccine and covid-19 vaccination in zimbabwe @zbcnewsoline @radiozim263 @healthtim	
002499(2)	2021	encouraging acceptance of covid-19 vaccine uptake @263chat @healthtimeszim @zbcnewsoline @infominzw	
002550(1)	2021	president mmangagwa received his first covid-19 vaccine in victoria fall	
002916(1)	2021	china pls donate vaccine to zimbabwe defence force defence and war veteran affair hon muchinguri-kashiri received a donation of do	
002952(1)	2021	@mohocczim virtual media sensitization meeting on typhoid conjugate vaccine tcv @mohocczim is introducing tcv into routine childhood	
002991(1)	2021	zimbabwe introduce typhoid conjugate vaccine @who_zimbabwe @unfpa_zimbabwe @263chat @healthtimeszim	
002992(1)	2021	vaccinate their children children are receiving their hpv vaccine at all school including chirrodzi primary	
003217(1)	2021	million war chest for vaccine	
003262(1)	2021	the ministry of health has mobilised vaccine for those in harare province in need of sinovac second dose those who are due for the	
003398(1)	2021	professor mthuli ncube we have about million vaccine that we have received so far going forward we have paid for another million v	
003431(2)	2021	kindly be advised that the two million covid-19 vaccine dose arrive on thursday july gabriel mugabe international airport add-hre	
003443(1)	2021	zimbabwe yesterday received million covid-19 vaccine from china sinovac at rgm international airport	
003445(1)	2021	current budget on vaccine acquisition is million cumulatively we have spent about million and what is left now is million if neces	
003447(1)	2021	harare july zimbabwe received million covid-19 vaccine from china sinovac at rgm international airport on july the roll-out plan i	
003450(3)	2021	track monitor the number of vaccinated people and vaccine used	
003575(1)	2021	today deputy minister dr john mangwiro toured central vaccine store to get an update of covid-19 vaccine stock status as of yester	
003582(2)	2021	covid-19 vaccine may be administered to people with underlying condition like cancer asthma hiv hypertension and diabetes liver and	
003695(3)	2021	dr john mangwiro received dose of the sinopharm vaccine at rgmia is good news for the country we currently have million dose of sin	
003738(1)	2021	today zimbabwe received sinovac vaccine zimbabwe total number of vaccine from purchase and donation is	
003885(1)	2021	sinopharm dose bring to the total number of vaccine donated by china to dose	
003941(2)	2021	globally billion vaccine dose have now been administered but percent have been administered in high and upper-middle income countr	
004058(2)	2022	#covax just delivered it billionth #covid19 vaccine dose grateful to all our partner and donor for their support and contribution	
004072(1)	2022	reply @who #covax the vaccine pillar of the @actaccelerator is leading the largest #covid19 vaccine procurement and supply operati	
004099(3)	2022	it remain more important than ever to get vaccine to the #vaccinequity	

Average and 95% confidence
Pos 1.5160 (1.4385, 1.5958)
Neg 1.8484 (1.7484, 1.9484)
Av pos - Av neg: -0.3324

Bootstrap 95% CIs
Pos 1.5160 (1.4402, 1.5977)
Neg 1.8484 (1.7551, 1.9475)

Geometric mean retweets/rate
(23.4357, 30.2201)

The above list are the first 38 matching posts to be listed on Covid-19 vaccine. The list in the above search gives the date that is 2020 to 2022 of the information sent and its positive and negative sentiment score on the far right upper corner. Matches were recognized by Mozdeh equaling the keyword 'Covid-19 vaccine' to all the texts and citing the texts containing the word. The search Covid-19 vaccine matches tweets containing covid or covid vaccine. For example, the first tweet shown in the search was sent in 2020 says, 'Zim preparing for covid-19 vaccine rollout...' The current analysis of tweets about the Covid-19 vaccine establishes that the number of tweets were updating people regarding the vaccine.

Sentiment analysis results:

Score	Pos	Neg
1	62.10%	45.77%

2	25.95%	30.61%
3	10.50%	17.20%
4	1.17%	5.83%
5	0.29%	0.58%

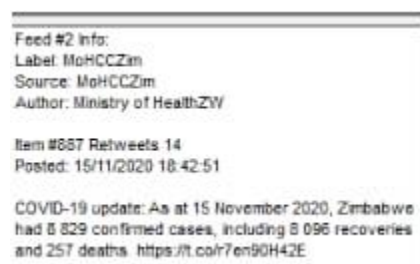
Averages: Relying from the above sentiment analysis results, when a search was running in Mozdeh, it premeditated the constructive and destructive sentiment strength of each corresponding text and reports the total average together with a 95% assurance interval, as shown in the results below.

Pos 1.5160 (1.4385, 1.5958)

Neg 1.7484 (1.7484, 1.9484)

From the results above, the negative sentiment is higher than positive sentiment and the difference is statistically significant. It is therefore be sensibly sure that people on Twitter tend to tweet more negative sentiment regarding covid-19 (vaccine).

The picture below is the information about Ministry of Health's tweets on the number of cases, recoveries and deaths displayed by Mozdeh when a search result was clicked. Specifically, it reveals the name of the author (Ministry of Health), retweet count, source, the tweet and date posted.



Evident from above, the Ministry of Health posted a tweet on the 15th of November 2020 about covid-19 update in the words, 'Covid-19 update: As at 15 November 2020, Zimbabwe had 5 829 confirmed cases, including 8 096 recoveries and 257 deaths...' The tweet had 14 retweets meaning people on Tweeter engaged on various tweets regarding Covid-19 (vaccines).

3.6 Conclusion

This chapter was concerned with analyzing the various strategies implemented by both anti-and pro-vaccination campaigners in order to disseminate the messages of the two antagonistic sides' impact on vaccination campaigns in Zimbabwe. Different communication strategies that exploited digital technology's multimodal techniques that combine the textual, pictorial/visual, sound and the performative to disseminate information utilised by social media users to spread Covid-19 vaccine information. The research also employed Social Media Indicator (SMI) and Mozdeh Software in illustrating the level of engagement of those people who were against the vaccine on social media and the extent they associate with their with other people on these media platforms and distinguish the main ideas from twits, or they can simply highpoint the topics of curiosity, and the hyperlinks among them respectively.

CHAPTER FOUR

HOW SOCIAL MEDIA PROMOTES VACCINATION CONSPIRACY THEORIES

4.0 Introduction

In this chapter the analysis of anti-vaccination and textual messages and images are analysed in search of how they help spread conspiracy theories and thus contributing to vaccine hesitancy. The theories end up creating myths which are very influential in persuading people to believe in certain issues despite their unproven basis. Because such myths operate in the context of social media they can be best described as fake news. The danger with such news is that they are propagandist in nature and audiences for such news in most cases fail to distinguish between fake and authentic news. Fake news plays an important role in disinformation and misinformation because of audiences' failure to make a distinction between factual and unfactual news. The techniques of deploying them such as the use of the comic are analysed to demonstrate how such techniques contribute in undermining factual issues and thus deceiving and misinforming the public. This chapter analyses social media

4.1 Social Media and Covid-19 Infodemic

The large volumes of messages communicated on social media and the problems of misinformation they generated resulted in an information pandemic that can be referred to as infodemic. Such a digital scenario happens when too much circulated information causes the misunderstanding rather than the understanding of a phenomenon. The Covid-19 vaccines information disseminated on social media platforms like WhatsApp, Twitter, Facebook and YouTube have influenced a lot of people in Zimbabwe to take informed or misinformed decisions regarding the safety of the vaccines currently been used for immunity. According to Duffy et al (2020: 8), 'A lot of people do not believe conspiracy theories regarding a Covid-19 vaccine, but a few people do, with belief specifically high among young people on social media.' Fake news about Covid-19 vaccines on social media platforms has instigated excessive levels of hesitancy among Zimbabweans. Interacting with Covid-19 vaccine messages on social media is

connected with undesirable perceptions and distrust in the effectiveness of the antidote among the population. After the outbreak of Covid-19 and thereafter, due to the advent of lockdown, people began to read of all kinds of news circulated on social media and it lay a ground for the development of fake news. Lazer et al. (2018) indicate that 'Fake news is invented data that mimics news media content inform but not in organizational process or intent.' Fake news on social media have a great impact on the public mainly because it (social media) has turn out to be more popular for news dissemination due to its easy access and low cost. This is a major reason of the circulation of fake news, specifically news with purposely false information that have destructive communal effects. Therefore, fake news on social media has become a research area that attracted great attention.

Digital media and particularly social media has been criticized as the main generator and spreader of fake news. Fake news can be defined as incorrect material deliberately created by a person with malicious agendas that misrepresent evidence, which is then diffused both intentionally and unintentionally by internet users. (Klein & Wueller, 2017:6) It is a term that was popularized by the former US president during the American presidential election of 2016 and throughout his presidential term because most media outlets were against him. Social media is the main culprit in generating and disseminating fake news because unlike conventional media, it is uncontrolled; individuals have the right and power to spread any messages they prefer regardless of its negative impact to others. There are no news editors like in conventional media to check the veracity of the messages. Another aspect is the rate at which social media spreads information. Social media users tend to share information without even verifying its authenticity by merely forwarding it to others. Thus fake information that could have started in one European country for instance, can reach every country in just a few hours or within a day depending on how fast people are sharing the information.

4.2 How Social Media Created Anti-Vaccination Myths and Conspiracy Theories

Myths and conspiracy theories about Covid-19 were not only with respect to vaccination but sanitization as well. The aim of conspiracy theorists was to ensure that if a person questions one measure to combat Covid-19, s/he would also question vaccination. The picture below was

circulated on social media to spread disinformation about some sanitization substances used to fight the Covid-19 pandemic.



The circulation of the picture above has a negative impact on individuals regarding sanitization and ultimately the uptake of the Covid-19 vaccine. The readers of such messages would think that the presumed negative effects of sanitization also translate to vaccination. The sender of this picture obviously intended to influence the public. Due to his or her words attached to the picture that say, 'Don't use sanitizer of this company. This is not a sanitizer. It is a new virus with early diseases. Share it with "everyone".' This picture reached a big population on the country's different social media platforms. The picture was received and forwarded by many people, for instance, the statement with black background was forwarded awakening people with the words, 'Pliz share *tibatsirike tese*' (... so that we help each other to spread the message), another sender attached the picture with the words, 'Guys pliz share.' The condition of the hands is revealing how deadly and toxic the sanitizer is, as well as the diction (choice of words) used by the image creator. For example, '...This is not a sanitizer. It is a new virus...' The statement implies that whatever is created to prevent the virus is creating another in human's body. Such fabricated information about important side effects of sanitizers caused many to hesitate sanitization.

The same digital technique was used with regard to vaccines. The following Zim Gazette's story about a Hwange District girl who died after being vaccinated made rounds on social media platforms.

📰 Victoria Falls Girl (12) Dies A Week After Vaccination
Health authorities in Hwange District are investigating the death of a Grade Seven pupil (12) who died after receiving the new typhoid conjugate vaccine (TCV) recently.

Ethel Tshuma from Village 8 in the Lubangwe area was vaccinated on 1 June and died on 8 June while admitted to St Patrick's Hospital in Hwange.

The family of the girl, who was a learner at Nyongolo Primary School outside Hwange, suspected she reacted to the jab as her body started swelling after she was vaccinated.

<https://www.zimgazette.co.zw/victoria-falls-girl-12-dies-a-week-after-vaccination/>

This kind of news instilled fear among the people of Zimbabwe especially those who have access to the internet. This piece of information is regarded as fake news because the only vaccine that was trending is that of covid-19, but knowingly the sender and editor chooses to post the heading with the statement, 'Victoria Falls Girl (12) Dies A Week After Vaccination'. Even though the vaccine that was referred to was not that of covid-19 but Typhoid Conjugate Vaccine (TCV), people began to inform others that it was covid-19 vaccine that killed the girl. Unintentionally people began to lie because of the diction that the editor used. Such misinformation and fake news have meted untold suffering to humankind and it is mental traumatizing and psychologically insipid. 'In recent times, social media has witnessed a reverberation amid the propagation of fake news which has made people hesitant to engage in honest news sharing for fear that such info is false' (Collins et al, 2020:1). Almost half of social media users sharing news reported having sometime shared misinformation. To be more frank, lack of transparent

and correct information about vaccine efficacy is a major reason that led to vaccine hesitancy in Zimbabwe.

4.2.1 Creation of Anti-Vaccination Myths

Myths are important in communication because even though they are not truthful, believers in myths see them as truthful. A myth can be defined as a popular belief or tradition that has grown up around something or someone. Generally in society myths are crucial because they validate traditions and values in society. According to Losada (2015:930);

A myth is an uttered story, symbolic, dynamic, that lacks any historical testimonial and it is composed of a series of constant or invariable cultural semantic elements, reducible to themes; and of a nature that is conflictive, functional and etiological.

Myth is basically visionary, an exaggeration of awareness into alternative realities. Due to the wide-spread anti-vaccination information circulated on different social media platforms as Zimbabwe was rolling out vaccines, the following 13 myths can be identified:

- Vaccines are unsafe because you die earlier when you get vaccinated
- The vaccine changes peoples' DNA
- Women will not get pregnant
- People with children are the ones to be vaccinated
- Covid-19 vaccines contain a form of microchip that will be used to control an individual
- Big businesses are pushing vaccines to improve profits
- Governments are complicit with big businesses in pushing vaccination programs despite the risks
- Another network is formulated - 5G
- You will be initiated into satanism after receiving a vaccine - 666
- Vaccines are a way for our former colonisers to colonise us again
- I do not need to get vaccinated if I already had Covid- 19
- It is politically related because the government will track me through the vaccine

- I will get infected by Covid-19 after receiving the vaccine
- Blacks cannot be infected

Big businesses are pushing vaccines to improve profits

Some people believed that Covid-19 is a business for doctors and they are diagnosing every flu-like diseases as Covid-19 for their own benefit. Ironically, in some places, people confronted funeral parlors for not handing over the wrong dead bodies. There was also a claim that Covid-19 was (is) an intentional project to insure the Bill Gates trackable microchip scheme. In the Zimbabwean context, people believed that the government provided false number of Covid-19 cases because a big number of cases will get more income and donation from different organisations and western countries.



Jay R. 4 months ago

Yes, every country with responsible leaders will be 'running the same script.' It's called common sense and the common good.

👍 7 🗨️ REPLY

Precisely, just like other countries, Zimbabwe also entered into direct talks with vaccine makers to ensure a timeous supply of vaccines. This is revealed on a YouTube comment posted by a YouTube user known as Jay R who regards all leaders that follow vaccination campaign as responsible.

Another network is formulated - 5G

The World Health Organisation (WHO) made it clear that viruses in this case (Covid-19) cannot travel on wireless waves and mobile networks. Several people in the country admitted to that myth because in the 21st century there has been a revolution in computing and communications and all indications are that technological progress will continue at an alarming pace. However, covid-19 spreads in many countries that do not have 5G mobile networks.

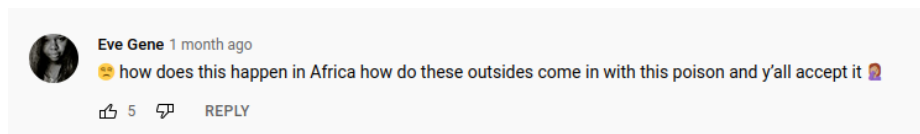
You will be initiated into Satanism after receiving a vaccine – 666

Other people especially Christians said they were not going to get vaccinated because the moment you are vaccinated you will be initiated into Satanism. They related to the advent of

covid-19 to the end of the world and they caricatured the event to what is written in the apocalyptic passage from Revelation that says Antichrist is going to trial Christians by making them receive a mark on their foreheads to earn a living. Some religious leaders spread conspiracy theories about vaccines, claimed that covid-19 vaccines herald the ‘mark of the beast’.

Vaccines are a way for our former colonisers to colonise us again

Anti-vaccine campaigners assume that the advent of covid-19 and its vaccines was (is) a way of the former coloniser to re-colonise the interior (Africa) again. A YouTuber spotted as Eve Gene commenting on a particular video on YouTube regards covid-19 vaccine as a poison intended to destroy Africans by the outsiders;



Several people began to quarrel among themselves why these vaccines were given for free. There was a myth that circulated during the covid times that, ‘Africa is going to be a colony of China by 2050’. By getting vaccinated people thought they were going to be used by the Chinese, the worse part of it, covid-19 originated in China. However, covid-19 vaccination campaign was not just the work of Western and rich countries but a global collaboration.

People with children are the ones to be vaccinated

There was a myth and rumor that circulated on social media platforms that people with children are the ones to get vaccinated. This is supported by a Facebook post on the screenshot below;



During covid times people used to believe most of the information that they read on social media, in this case, a myth in the above picture perceptibly impacted a number of people in the

vaccination uptake. A persona by the username Magolide Da Vocalist wrote a Face book post that says, ‘Vaccine hanzi yobaiwa vanhu vane vana isusu tisina hanzi hatibaiwe’ (The vaccine is for people with children, we are not included’ This statement is short but powerful in terms of its effect that it impose among people regarding the vaccine uptake.

Women will not get pregnant after receiving a dose

The core of anti-vaccine belief is a conspiracy theory that vaccines do not function and are damaging because they can cause infertility among teenaged girls. This myth implanted fear among young generation especially girls because they were afraid of living a childless life.

Vaccines are unsafe because you die earlier when you get vaccinated

Several people were afraid of getting vaccinated because of a rumor that says, If you get vaccinated you will die after two (2) years. The only way to escape death was to abstain from getting vaccinated.

4.3 Role Played by Comic Messages on Social Media Regarding Vaccines

During the first year of the Covid-19, people in Zimbabwe produced and shared hundreds of humorous Covid-19 vaccine jokes in form of memes (written, audio and visual) via different social media platforms. Several jokes were about the little problems of the vaccine, not about the virus per se, or about sick people. ‘This is why Covid-19 led to an increase in mental health problems such as stress, anxiety, anger, and insomnia’ (Torales et al. 2020). A major way adopted by people in dealing with the situation as mentioned above by Torales and his crew, was by producing and sharing Covid-19 or vaccine jokes and memes. Humor is significant because of its capacity to boost message understanding and acceptance. Funny messages and media circulated on social media also poses negative impact on people’s reaction regarding vaccines. Therefore, this section extends research on the effectiveness of humorous messages imbedded in media intended for social media engagement.

In Zimbabwe funny or comic messages became popular during the Covid-19 pandemic. Sharing jokes has become the common form of communication over social media platforms, as indicated

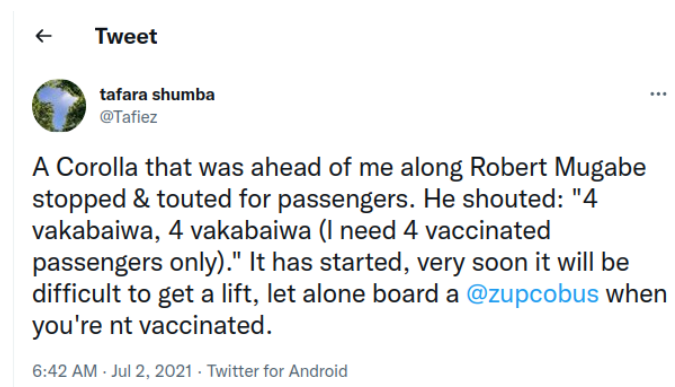
by the unprecedented number of memes shared everyday as well as traditional social media, for instance, Facebook, WhatsApp, Twitter among the other. When viewed in the context of social media, funny messages and media are seen as a welcoming entertainment from the content clutter over-whelming social media audiences. Social media users not only spectators but also engage in (re-) sharing the vaccine funny messages or media posted by their contacts, often with a larger audience. A screenshot below was taken on Facebook, is a particular post about Covid-19 vaccine sent by a user with the name Mwana Wamai Dhor.



Drawing evidence from the above screenshot, propagating funny messages and laughing together helps people to share experiences and collectively make sense of what is going on (Tracy et al. 2006). Precisely, the sender is informing the public that a lot of people are getting vaccinated so he is inspiring others to follow suit. According to him, at first, people listened to conspiracy theories and myths regarding the vaccine and inspired each other not to get vaccinated. This is witnessed by the strong use of the Shona borrowed word *madhiri* in the statement, ‘...*uchida kuona kuti munhu wenyama haaite kurongeka naye madhiri check tese pataiti hatibaiwe vaccine...*’ (People of this world are betrayers, when we said we were not going to get vaccinated). The mentioned joke explores that Covid-19 vaccine is just like other vaccines that were introduced in the country. The same point is being revealed by the second joke that is implying that, at first people hated the vaccine but as time moves they started to understand it

and demystify certain myths regarding Covid-19 vaccine. Social media jokes regarding vaccines were not mere jokes. However, despite the positive aspects of humour it also has its negative consequences in communication. In serious matters such as Covid-19, it tends to water down the importance of the message making readers see the message as trivial.

The Zimbabwean government declared numerous tight measures or rules to limit the spread of the Covid-19 by increasing herd immunity. Some of these rules included, no job-no vaccine, no travelling without getting vaccinated, vaccine card-personal passport, to get entry into the church one should have been vaccinated to mention but a few. These measures also gave rise to different kinds of comic messages and media on various social media platforms and they were received by hundreds of people because jokes grab attention and stimulate a person's thought. Relying on this concept, the researcher predicts that moving messages are especially attractive to people in times of existential crisis, as they become more meaningful (Oliver and Raney 2011). The screenshot underneath is from a Twitter handle, a user known as tafara shumba posted funny message of how mushikashika drivers call for passengers when Covid-19 was at its peak in 2021 by mentioning the issue of vaccines.



Within humorous funny messages and media, the population began to exchange more serious and optimistic, moving messages during the Covid-19 crisis. In reality, mushikashika used to carry both the vaccinated and unvaccinated passengers regard their vaccine status. However, the driver touted with the statement, '4 vakabaiwa, 4 vakabaiwa' (I need four (4) vaccinated passengers only). He was unintentionally making known to people one among the regulations of travelling during country's national lockdown. The person tend to use humor, in this case, funny

message to attract other people's attention and to inform them about the importance of covid-19 vaccine.

4.4 Conclusion

This chapter analysed social media message in terms of how they contributed to the creation of myths or fake news and falsehoods that supported conspiracy theories regarding vaccination. . These theories ended up generating myths which were very influential in persuading people to believe in certain issues despite their unconfirmed basis. Fake news and myths regarding Covid-19 vaccine play an important role in disinformation and misinformation because of audiences' failure to make a distinction between factual and unfactual news. Myths and conspiracy theories about Covid-19 were not only with respect to vaccination but other ways implemented by the government to combat the virus such as sanitization.

CHAPTER FIVE

CONCLUSION

5.0 Introduction

The study was concerned with analysing the impact of pro- and anti-Covid-19 vaccination messages posted on social media in promoting vaccine hesitancy in Zimbabwe since the introduction of Covid-19 vaccines. The previous chapters investigated the background to the emergence and spread of the Covid-19 pandemic in order to demonstrate why vaccines were necessary as one of the strategies adopted to fight the pandemic locally and globally. The study explicated how social media contributed to the problem of Covid-19 vaccine hesitancy in Zimbabwe in order to establish how digital media is be used as a propaganda campaign against strategies to combat the highly lethal coronavirus. The research drawn special attention to the effect, specifically Covid-19 vaccine hesitancy, imposed by social media (Twitter, Facebook, WhatsApp and YouTube) messages. It also established the ways in which digital communication in the form of social media is being used for wrong reasons in spite of its benefits to society in form of fake news, myths and conspiracy theories. The chapter concludes by presenting the research findings, recommendations for future practice and for further research.

5.1 Research findings

The study has highlighted that the low Covid-19 vaccination turnout has been associated with distortions circulated on basic social media platforms, thereby triggering vaccine hesitancy. Some people were not in a hurry to get vaccinated until they fell sick with Covid-19. People believed in misinformation and conspiracy theories about vaccines. The theories ended up creating myths which were very influential in persuading people to believe in certain issues despite their unconfirmed basis. Because such myths operate in the context of social media they can be best described as fake news. Fake news played an imperative role in disinformation and

misinformation because of audiences' failure to make a distinction between accurate and inaccurate news.

This study thus accessed social media information and analysed it and exposed why it was so influential in persuading many people against vaccination. Social media platforms were utilised to obtain updates on the Covid-19 epidemic more as compared to conventional sources of information like television, radio, and newspapers. However, instead of utilizing the internet, to be in particular, social media for the benefit of the society people began to misuse it by demotivating others when it comes to Covid-19 vaccination. Some people spread false information and news intentionally regarding the Covid-19 pandemic *per se*. The study discovers that, users that create content on social media are inspired by factors that include peer interrelation, popularity seeking, notoriety, and most importantly self-establishment. Not all messages were favourable, some information was against the vaccine and it instilled fear among people thereby triggering vaccine hesitancy.

The research established the various strategies adopted by both anti-and pro-vaccination campaigners and revealed the messages of the two antagonistic sides' impact on vaccination campaigns in Zimbabwe. Anti-vaccination promoted vaccine hesitancy as compared to the other that was pro-vaccination. The two types of messages resulted in two frames of meaning that were witnessed on different social media platforms during the ravaging Covid-19 pandemic. With studies from 2020 until today, large quantities of the framed information about vaccines on prevalent social media platforms like Facebook, WhatsApp, Twitter to mention but a few, are anti-vaccination messages. There was a strong anti-vaccination movement in Zimbabwe that was witnessed on various framed messages on social media. This research explored that message frames play an immense role in Covid-19 vaccination attitudes and behavior among people. Social media information that contains forceful language(s) increased psychological reactance, which significantly weaken the effect of persuasive.

The study also established the extent to which the media spectacle promoted Covid-19 vaccination and undermined its intended objective. It also analysed how Covid-19 vaccine was captured and mediated to a watching world specifically, the nature of debates surrounding the Covid-19 vaccine on social media. Vaccine campaign and news in Zimbabwe have all been

subjected to the logic of spectacle in the era of media sensationalism. During Covid-19 times certain debates were progressively engaged on the screens of media culture, which displays spectacles like the Covid-19 vaccines. The media spectacle regarding the coronavirus vaccination indicated during the research is that the constant and continued spectacularisation of vaccination was reacting against disinformation and misinformation from anti-vaccination and Covid-19 denialists which had a huge following.

The study also established that the various explanations behind vaccine hesitancy, religious issue is one of the vital reasons specifically, the Apostolic sector. The study indicated that Independent African Churches were another section of the population that is against the use of western medicine or the seeking of health solutions at modern health facilities. Besides religious and traditional factors promoting vaccine hesitancy there are also socio-economic factors. In spite of the religious reasons to Covid-19 vaccine doubtfulness, this study also clarified that, a greater vaccine hesitancy was among the less-educated, females, ethnic minorities, people living in rural areas, and those with lower income.

5.2 Recommendations

In light of the above research findings, the study offers recommendations for future studies on the effect of social media and Covid-19 vaccine hesitancy on vaccine uptake.

5.2.1 Recommendations for future research

The study recommends:

1. The study only concentrated on social media ‘big data’ to address its research problem. For more accurate results social media data could have been completed by more data gathered from propounders of conspiracy theorists, leaders of churches against vaccination and other relevant players in vaccination campaigns.

2. Studies on how people can be educated to use social media responsibly are necessary so that social media can influence society positively on the basis of its huge power to shape public opinion.
3. The study did not cover all social media platforms in its investigation of vaccine hesitancy. More research could be done incorporating other digital platforms such as personal blogs and, Tik Tok.

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